

GE Healthcare

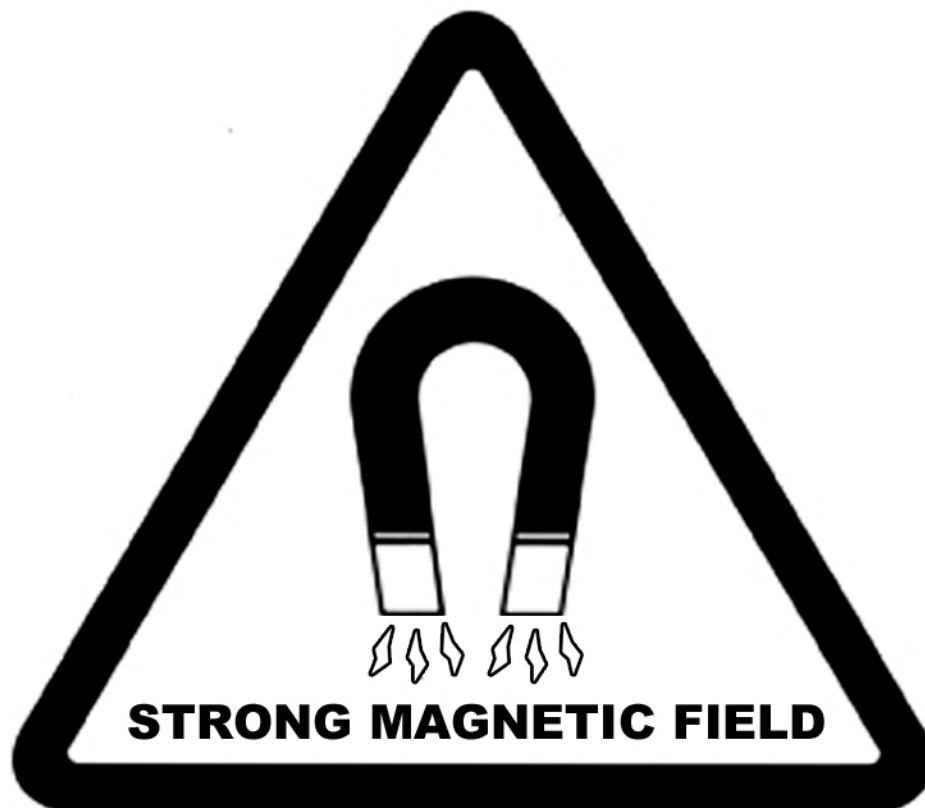
3.0T LCC300 Magnet Delivery & Installation



OPERATING DOCUMENTATION

5500098
Revision 4

WARNING



NO PACEMAKERS NO METALLIC IMPLANTS

Persons with pacemakers, neurostimulators or metallic implants must not enter the magnet area. Serious injury may result.



NO LOOSE METAL OBJECTS

Iron and steel materials must not be taken into the magnet area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ

(BG)

Това упътване за работа е налично само на английски език с изключението на случаите, когато се изисква изрично по друг начин от местното законодателство, или от споразумение на местно ниво.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告

(ZH-CN)

本维修手册仅提供英文版本，除非当地法律或本地供应协议另有明确要求。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

Tento provozní návod existuje pouze v anglickém jazyce, pokud není jinak výslovně vyžadováno místními zákony nebo odsouhlaseno na místní úrovni.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

Denne servicemanual findes kun på engelsk, medmindre andet kræves i henhold til lokal lovgivning eller lokal aftale.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar, tenzij expliciet vereist door plaatselijke regelgeving of overeengekomen op lokaal niveau.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

This service manual is available in english only except as otherwise expressly required by local law or agreed to at a local level.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS

(ET)

See teenindusjuhend on saadaval ainult inglise keeles, kui kohalikud seadused ei ütle teisiti või kui kohalikes õigusaktides ei ole otseselt teisiti ette nähtud.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

Tämä huolto-ohje on saatavilla vain englanniksi, ellei paikallinen laki nimenomaan toisin vaadi tai jos toisin on sovittu paikallisella tasolla.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

Sauf exigence contraire des lois locales ou accord contraire au niveau local, ce manuel d'installation et de maintenance n'est disponible qu'en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

Diese Serviceanleitung existiert nur in englischer Sprache, sofern nichts anderes gesetzlich vorgeschrieben oder auf lokaler Ebene vereinbart wurde.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά, εκτός αν η τοπική νομοθεσία απαιτεί κάτι άλλο ή υπάρχει διαφορετική συμφωνία σε τοπικό επίπεδο.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el, kivéve ha a helyi rendelkezések ezt kifejezetten elő nem írják, illetve ha helyi szinten erről külön megállapodás nem születik.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

Þessi þjónustuhandbók er eingöngu fánleg á ensku, nema annað sé sérstaklega krafist, löglega eða samþykkt á landsgrundvelli.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

Il presente manuale di manutenzione è disponibile soltanto in inglese, eccetto quando espressamente richiesto dalle normative locali o convenuto a livello locale.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告

(JA)

このサービスマニュアルには英語版しかありません。ただし使用国の法令に別異の定めがある、あるいは現地で別段の合意がある場合を除きます。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

현지 법률에 따라 명시적으로 요구하거나 현지 수준에서 합의한 경우를 제외하고 본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS

(LV)

Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā, izņemot gadījumus, kad vietējie likumi nepārprotami nosaka citādi vai panākta vienošanās vietējā līmenī.

- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

Šis eksploatavimo vadovas yra tik anglų kalba, išskyrus tuos atvejus, kai vietiniai įstatymai tiesiogiai numato kitokius reikalavimus arba vietinių lygiu sutarta kitaip.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL

(NO)

Denne servicehåndboken finnes bare på engelsk, bortsett fra dersom det motsatte uttrykkelig er fastsatt av lokal lovgivning eller det er inngått annen avtale lokalt.

- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim, chyba że lokalne przepisy lub umowy wyraźnie stanowią inaczej.

- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

ATENȚIE

(RO)

Acest manual de service este disponibil numai în limba engleză, cu excepția cazului în care este o cerință obligatorie stipulată de legislația națională sau convenită la nivel local.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

Данное руководство по техническому обслуживанию предлагается только на английском языке, за исключением тех случаев, когда наличие руководства на национальном языке является требованием местного законодательства или когда выпуск такого руководства согласован с местным представительством.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

Tento návod na obsluhu je k dispozícii len v angličtine, okrem prípadov, kedy tak výslovne vyžadujú miestne zákony alebo je dohodnuté na miestnej úrovni.

- Ak zákaznikov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION

(ES)

Este manual de servicio sólo existe en inglés, salvo que la legislación local exija de forma expresa lo contrario, o así se haya acordado a nivel local.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

Den här servicehandboken finns bara tillgänglig på engelska om inte annat uttryckligen krävs av lokal lag eller har överenskommit på lokal nivå.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT

(TR)

Aksi, yerel bir yasa tarafından açıkça gerekli görülmediği veya yerel bir seviyede kabul edilmediği takdirde, bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

AVISO

(PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês, salvo disposições em contrário previstas pela legislação local ou acordadas no âmbito local.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO
(PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês, salvo requisição expressa pela legislação local ou acordo efectuado a nível local.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

UPOZORENJE
(SR)

Ovo servisno uputstvo je dostupno samo na engleskom jeziku, sem ako lokalni zakon to izričito zahteva ili je dogovoreno na lokalnom nivou.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORENJE
(HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

警告
(ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

警告
(ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

OPOZORILO (SL)

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

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Revision History

Rev	Date	Primary Reason for Change
4	Jan. 29, 2010	Per eNPI 012683: Changed “HDv” to MR750 where applicable. Per DOC0645958: Updated the first NOTICE in Section 1.2 Electrical Checks in Chapter 7, Magnet System Checks (by GE Service). Per ECO2092367: Updated Chapter 6, International (Air-Sea) Magnet Delivery, with the new style of crate and pallet. Per ECO2091912: Added a note that aluminum spacers must remain on magnet during movement of magnet in Chapter 9, Moving Magnet to MR Suite. Per SPR MCIhc47704: Updated Chapter 4, Section 1.2, GE PMI and Shield Room Vendor. Updated Chapter 7, Section 1.1 Physical Inspection. Updated Chapter 8, Section 1 Surface Mounted VibroAcoustic Damping Mate Placement. Updated Chapter 9, Moving Magnet to MR Suite. Updated Chapter 10 - Magnet Leveling, Foot Shimming and Bolt.
3	June 22, 2009	Per ECO 2059126: Updated the illustration for Positioning VibroAcoustic Damping Mats with measurements for positioning the 8 in. Quench Vent.
2	Dec. 2, 2008	Per ECO 2059126: Made changes to Chapters 4, 8, 9 and 10 for surface-mounted VibroAcoustic Damping Mats and aluminum spacers issue. Per ECR 2068697: Changed the title of this manual by removing “Signa”. Per PQR 13224527: Added a Note to Step 2 in Chapter 10, Magnet Leveling, Foot Shimming & Bolt Down.
1	Apr. 30, 2008	Initial release. A new manual was created for HDi, HDx and MR750 to satisfy vibroacoustic damping mat issue in PQR 13101624.

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Chapter 1 Damage In Transportation

1 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

Rev. June 13, 2006

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Chapter 2 Safety Alert Usage

1 Safety Alert Usage

The following ANSI (American National Standards Institute) Z535.4 definitions are applied to notify users of the potential for **personal** injury:



DANGER

INDICATES AN IMMINENTLY HAZARDOUS SITUATION THAT, IF NOT AVOIDED, WILL RESULT IN DEATH OR SERIOUS INJURY. DANGER IS LIMITED TO EXTREME SITUATIONS.



WARNING

INDICATES A POTENTIAL HAZARDOUS SITUATION THAT, IF NOT AVOIDED, COULD RESULT IN DEATH OR SERIOUS INJURY.



CAUTION

Indicates a potential hazardous situation that, if not avoided, may result in minor or moderate injury.

The following ANSI Z535.2 definition is applied to notify users of the potential for property damage only:



NOTICE

Indicates information or company policy that relates directly or indirectly to the safety of personnel or protection of property.

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Chapter 3 Overview

1 Overview

This Signa 3.0T LCC300 Magnet Delivery & Installation Manual addresses moving, delivery and installation of the 3.0T magnet at its location in the MR Suite.

The sequence of events involved in magnet delivery and installation, along with functions, responsibilities, and associated documentation are shown below.

Table 3-1: Magnet Delivery and Installation Functions and Responsibilities

	Function	Responsibility	Reference Document
1	Site Delivery Review: - Access and Route - Clearances	GE Project Manager, Installation (PMI), GE Field Engineer and Rigger	Appropriate Pre-Installation Manual and this Delivery and Installation manual.
2	Magnet Delivery	Rigger	Chapter 5, Domestic Magnet Delivery and Chapter 6, International (Air-Sea) Magnet Delivery in this manual.
3	Magnet Component Checks: - Physical - Electrical	GE Field Engineer	Chapter 7, Magnet System Checks (by GE Service) in this manual.
4	Place VibroAcoustic Damping Mats (Sites receiving M1060MA)	Rigger	Chapter 8, VibroAcoustic Damping Mat Placement in this manual.
5	Move Magnet to MR Suite	Rigger	Chapter 9, Moving Magnet to MR Suite in this manual.
6	Magnet Leveling & Bolt Down	Rigger	Chapter 10, Magnet Leveling, Foot Shimming and Bolt Down in this manual.
7	MR Magnet Safety Document Review & Compliance	GE Field Engineer	2301164PRE, <i>MR Magnet - Safety Requirements</i>
8	Magnet Conversion to Operating Configuration and Exhaust Gas Vent Connection Within 24 Hours	GE Field Engineer	Set-Up and Calibration chapter in the following manuals: - For HDi and HDx: Direction 2325852, <i>LCC300 Active Shield Magnet and Cryogens Subsystem</i> - For MR750: Direction 5500099, <i>MR750 3.0T LCC300 Active Shield Magnet and Cryogens Subsystem</i>
9	Continue with Magnet Commissioning in the Set-Up and Calibration chapter in: - For HDi and HDx: Direction 2325852, <i>LCC300 Active Shield Magnet and Cryogens Subsystem</i> - For MR750: Direction 5500099, <i>MR750 3.0T LCC300 Active Shield Magnet and Cryogens Subsystem</i>		

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Chapter 4 Pre-Delivery Instructions

1 Pre-Delivery Instructions

1.1 GE Project Manager of Installation (PMI) and Rigger



NOTICE

Rigger is responsible for actual equipment/procedures used to lift and move a magnet into the customer facility, including through a raised opening in an exterior wall. The following EXAMPLE procedure only outlines the concept of one method.



NOTICE

The magnet ships with some enclosures installed. Pushing on the magnet enclosures can damage the enclosures and potentially the magnet.

Before magnet delivery, the GE PMI and Rigger **must**:

1. Visit the magnet site with the rigging foreman before magnet delivery to plan the move. Hand deliver a copy of Direction 5500098, *3.0T LCC300 Magnet Delivery & Installation* to the rigging foreman. Review the guidelines for shipping/handling and for equipment/tools in [Chapter 6, International \(Air-Sea\) Magnet Delivery](#) (for crated magnet shipments) and in [Chapter 5, Domestic Magnet Delivery](#) (for all magnet shipments) with the rigging foreman.
2. Caution the rigger that the magnet is extremely fragile. Sudden jolts can damage the magnet. (Make riggers aware of the cost of a magnet replacement. This usually promotes more care while handling the magnet.)
3. Make sure all roads and paths leading to the magnet room are level and free from obstacles and holes. (The rigger will be required to construct platforms where needed.)
4. Verify clearance dimensions along the path to the magnet room meet the requirements stated in the appropriate Pre-Installation manual. In the case of a magnet being lifted by a crane through a raised opening, make sure the opening is **at least** 96 in. (2,440 mm) wide and 102 in. (2,590 mm) tall.
5. If roller dollies are to be used, have the rigger bring steel plates to place along the delivery route. Refer to the Miscellaneous Equipment/Tools subsection of [Chapter 5, Domestic Magnet Delivery](#) for rigger-supplied materials.
6. Floors must be protected using necessary processes.

1.2 GE PMI and Shield Room Vendor

Before magnet delivery, the GE PMI and Shield Room Vendor **must**:

1. Verify that floor levelness specifications stated in the appropriate Pre-Installation manual are met after the finished floor is installed.
2. Mark on the magnet room floor where VibroAcoustic Damping Mats will be located if the site is getting VibroAcoustic Damping Option A.

NOTE: VibroAcoustic Damping Option A provides VibroAcoustic Damping Mats for initial installation. See [Chapter 8, VibroAcoustic Damping Mat Placement](#).

**NOTICE**

Make sure all equipment anchors are located in conformance with the site's architectural drawings and installed and tested per the *Anchor Hardware Requirements for MR Equipment inside RF Shield Room* section of the appropriate Pre-Installation manual. Improperly located anchors can cause image quality issues.

3. Verify that the vent is located according to the specifications stated in the appropriate Pre-Installation manual.

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Chapter 5 Domestic Magnet Delivery

1 Domestic Magnet Delivery



WARNING

POTENTIAL ASPHYXIATION HAZARD.
LOSS OF MAGNET VACUUM WILL RESULT IN THE RAPID EXPULSION OF
HELIUM GAS, WHICH CAN CAUSE ASPHYXIATION IN ENCLOSED AREAS.

USE EXTREME CAUTION TO NOT CONTACT OR DAMAGE THE VACUUM
VESSEL DURING MAGNET TRANSIT OR SITING.

1.1 Equipment and Tools for Delivery & Installation

1.1.1 Forklift

Table 5-1: Forklift Requirements

Item	Qty.	Equipment Specification/Rating		Furnished By	Intended Use
Forklift	1	Forklift Capacity:	HDi and HDx: 24,810 lbs. (11,254 kg) at 50 in. (1,270 mm) to load center, minimum MR750: 25,709 lbs. (11,686 kg) at 50 in. (1,270 mm) to load center, minimum	Rigger	Unloading or moving Magnet (without pallet)
		Minimum Fork Length:	90 in. (2,286 mm)		
		Minimum Distance Between Forks using Lifting Rails:	80 in. (2,032 mm)		

1.1.2 Crane Lift



WARNING

POTENTIAL INJURY HAZARD
FAILURE OF EQUIPMENT/TOOLS USED DURING CRANE LIFTING MAY
RESULT IN SERIOUS PERSONAL INJURY.

ALWAYS USE LIFTING APPARATUS (CRANE, SPREADER BEAM, ETC.)
MEETING THE SPECIFICATIONS STATED IN THE TABLE BELOW.

Table 5-2: Crane Requirements

Item	Qty.	Equipment Specification/Rating		Furnished By	Intended Use
Crane	1	Total Load:	HDi and HDx: 24,810 lbs. (11,254 kg), minimum MR750: 25,709 lbs. (11,686 kg), minimum	Rigger	Unloading or moving Magnet (without pallet)

Item	Qty.	Equipment Specification/Rating		Furnished By	Intended Use
Spreader Beam	1	Total Load:	HDi and HDx: 24,810 lbs. (11,254 kg), minimum MR750: 25,709 lbs. (11,686 kg), minimum		
		Distance Between Lifting Points, Underside of Spreader Beam	90 to 96 in. (2,295 to 2,438 mm)		
Two-Legged Cable Bridle/Sling (Lifting Rail to Spreader Beam)	2	Total Load:	HDi and HDx: 24,810 lbs. (11,254 kg), minimum MR750: 25,709 lbs. (11,686 kg), minimum		
		Leg Length:	90 in. (2,286 mm) min. (all legs same length)		
		Load Per Leg:	HDi and HDx: 12,405 lbs. (5,627 kg), minimum MR750: 12,854.5 lbs. (5,843 kg)		
Two-Legged Cable Bridle/Sling (Spreader Beam to Crane)	2	Leg Length:	90 in. (2,286 mm) min. (all legs same length)		
		Load Per Leg:	HDi and HDx: 24,810 lbs. (11,254 kg), minimum MR750: 25,709 lbs. (11,686 kg), minimum		
Shackle	4	Pin Diameter:	1 in. (25 mm)		
		Minimum Workload Limit:	HDi and HDx: 12,405 lbs. (5,627 kg), minimum MR750: 12,854.5 lbs. (5,843 kg)		
Chain Hoist	2	Maximum Length:	90 in. (2,286 mm) Actual length required to be determined by rigger.		
		Minimum Workload Limit:	HDi and HDx: 12,405 lbs. (5,627 kg), minimum MR750: 12,854.5 lbs. (5,843 kg)		



WARNING

POTENTIAL INJURY HAZARD.
MAGNET IS AN UNBALANCED LOAD. TILTING CAN RESULT IN A HAZARDOUS CONDITION THAT CAN RESULT IN MAGNET DAMAGE OR SERIOUS PERSONAL INJURY.

TO PREVENT TILTING:

- MAKE SURE THE LIFTING APPARATUS (CRANE, SPREADER BEAM, ETC.) MEETS THE SPECIFICATIONS STATED IN [TABLE 5-2](#) AND [ILLUSTRATION 5-1](#).
- ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.
- ADJUST LIFTING CABLES/SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF THE SURFACE.

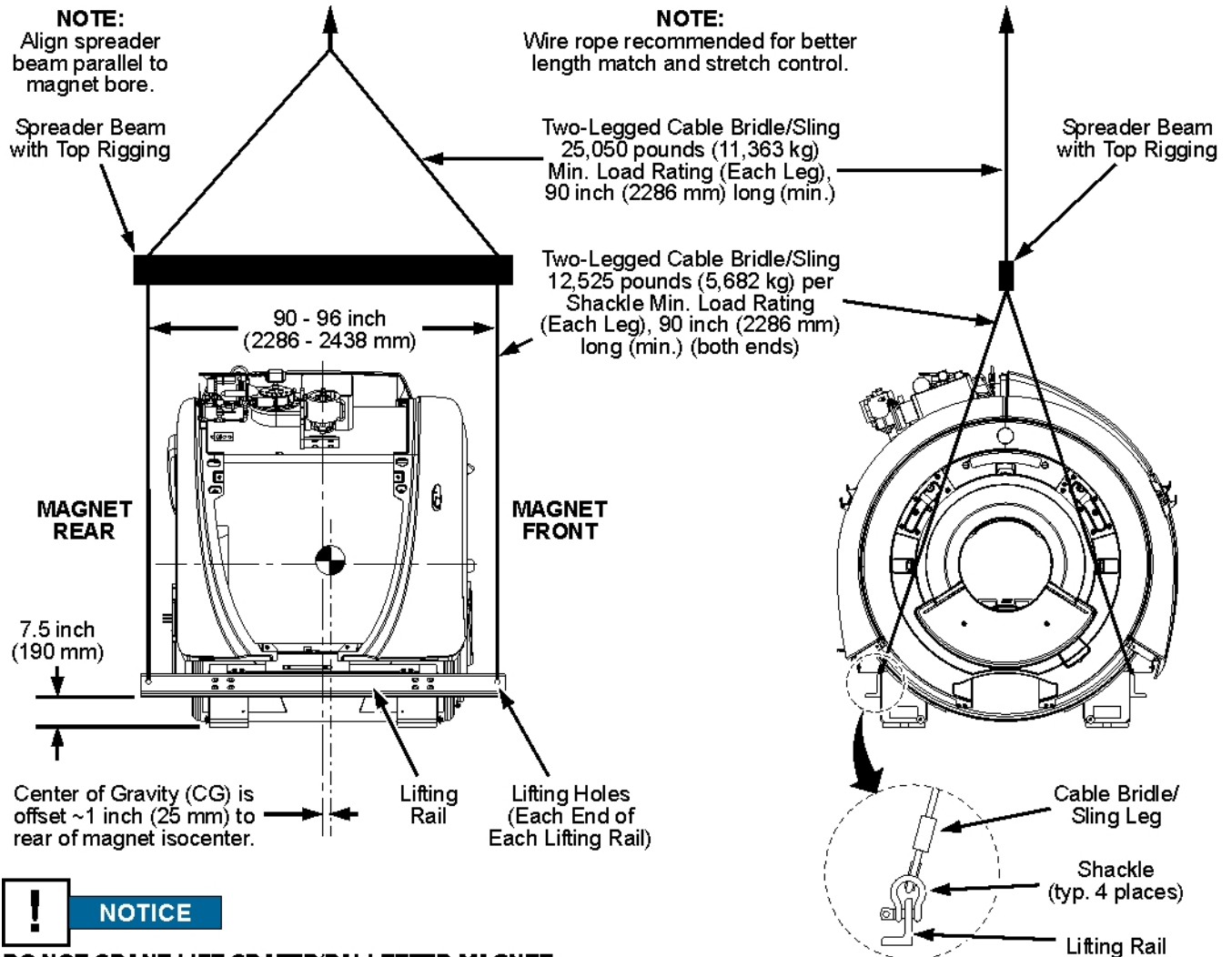
**WARNING****POTENTIAL INJURY HAZARDS**

TO PREVENT LIFTING CONDITIONS THAT MAY BE HAZARDOUS TO PERSONNEL AND DAMAGING TO THE MAGNET AND EQUIPMENT:

- MAKE SURE THE ENTIRE AREA WHERE LIFTING WILL OCCUR IS FREE OF OBSTRUCTIONS AND UNAUTHORIZED PERSONNEL.
- MAKE SURE THE SURFACE WHERE THE MAGNET WILL BE PLACED AFTER LIFTING IS FLAT.
- DO NOT CRANE LIFT A MAGNET DURING DANGEROUS WEATHER CONDITIONS.

NOTE: **Difference for MR750:** The maximum load for the Two-Legged Bridle Sling is 25,709 lbs. (11,686 kg) as shown in Illustration 1.

Illustration 5-1: Crane Lift Configuration

**NOTICE**

DO NOT CRANE LIFT CRATED/PALLETED MAGNET.
Crated/palletted magnets must be handled in conformance with the International (Air/Sea) Magnet Delivery section of this manual.

1.1.3 Miscellaneous Equipment and Tools

Table 5-3: Miscellaneous Equipment and Tools

Item	Equipment/Tool Requirement	Responsible	Function
Hydraulic or Toe Jack	Must support one end of magnet on two jacks or both ends of magnet on 4 jacks. Jack capacity must be greater than 12,525 lbs. (5,682 kg) per jack.	Rigger	Raise magnet for roller dollies or leveling plates.
Roller Dollies	Jack capacity must be greater than 12,525 lbs. (5,682 kg) per jack.		Moving magnet to magnet room.
Level	24 to 36 in. (610 to 915 mm) length		Level the magnet.
Magnet Leveling Kit	46-260888G4		Level the magnet.

Item	Equipment/Tool Requirement	Responsible	Function
Magnet Boltdown	To secure the magnet to the vibromat: Minimum 250 ft-lbs (339 Nm). If vibromat is not used, refer to anchor specifications per the anchor vendor.	Shield Room Vendor	Bolt down magnet.
Torque Wrench and Socket	Tighten the bolts on the magnet to achieve 250 ft-lbs (339 Nm). Deep socket size is dependant on anchor selection.	Rigger	Bolt down magnet.
Socket Wrench and Sockets	1/2 and 3/4 in. sockets		Remove 1/2 in. crate lag screws.
Magnet Interface Drawing	5171937IDW	GE Installation Specialist	Identify magnet dimensions and features.
Pre-Delivery Information Package	N/A	GE	Aid in magnet installation.
Portable Temperature Meter	2125073	GE Field Engineer	Check internal temperatures.
Digital Volt/Ohm Meter	Fluke model or equivalent.		Perform magnet electrical checks.
Chapter 7, Magnet System Checks (by GE Service)	N/A		Confirm electrical check values.
Wood Timber	As needed.	Rigger	Moving/positioning magnet to magnet room.
Steel Plate			Moving magnet to magnet room.
Pry Bar	Length as required to move magnet.		Moving/positioning magnet to magnet room.
RuO Temperature Meter	2171219	GE	Check Coldhead second stage and recondenser temperatures.
Adapter Cable for Diode Temperature Meter	N/A		Check internal temperatures.
Circular or Chain Saw		Rigger	For cutting wood timbers.
Heavy Duty Drill and Drill Bits			To add holes to wood timbers.
Minimum 0.50 in. (12.7 mm) Diameter Mounting Hardware			For attaching wood timbers to rails.

1.2 In-Transit Service



NOTICE

In-transit service **MUST** be performed by qualified personnel only and in **STRICT** conformance with the Liquid Helium Fill procedure stated in the appropriate manual:

- Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem* OR
- Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem* AND
- *2301164PRE MR Magnet Safety Requirements*, which is included with both of the above manuals.

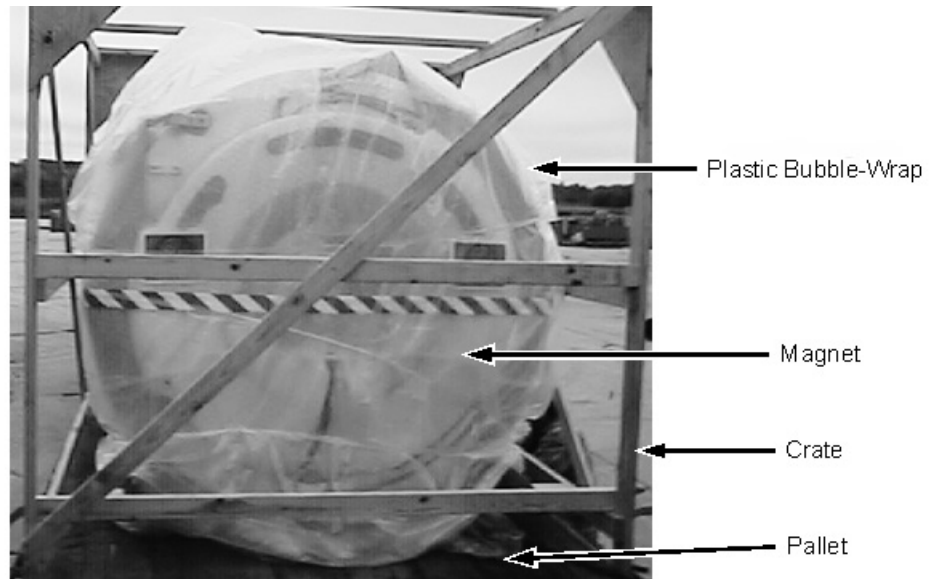
These documents are available through the Common Documentation Library at gehealthcare.com or your GE Healthcare Field Service Representative.

NOTE: In-transit helium refill is performed based upon the magnet shipping date per the documents stated above.

In-transit electrical checks are covered in [Chapter 7, Magnet System Checks \(by GE Service\)](#).

1.3 Removal of Shipping Cage and Protective Packaging

Illustration 5-2: Domestic Shipping Configuration



NOTICE

Care must be taken not to scrape or hit the sides of the magnet.

The magnet is shipped inside plastic bubble-wrap. The bubble-wrap should be left intact until the magnet is anchored in the magnet room.

Remove the shipping cage from the magnet using a crane. Either lift the cage straight up or unbolt the boards on one end of the cage, and move the cage away from the magnet in the other direction.

1.4 Unloading the Magnet



WARNING

POTENTIAL INJURY HAZARD.
PUSHING MAGNET ENCLOSURES MAY RESULT IN BODILY INJURY TO PERSONNEL.

DO NOT PUSH MAGNET ENCLOSURES.



NOTICE

To prevent magnet damage:

- Do not apply any force to the magnet enclosures. (See [Illustration 5-3](#).)
- Only use equipment/tools that meet the specifications stated in [Table 5-1](#) to fork lift the magnet off the shipping pallet or in [Table 5-2](#) to crane lift the magnet.
- Before moving the magnet into the MR Suite, refer to [Chapter 9, Moving Magnet to MR Suite](#).

Illustration 5-3: Do Not Push Signs



One on Left Side
One on Right Side



Two on Front End
Two on Rear End



1.4.1 Unloading with a Forklift

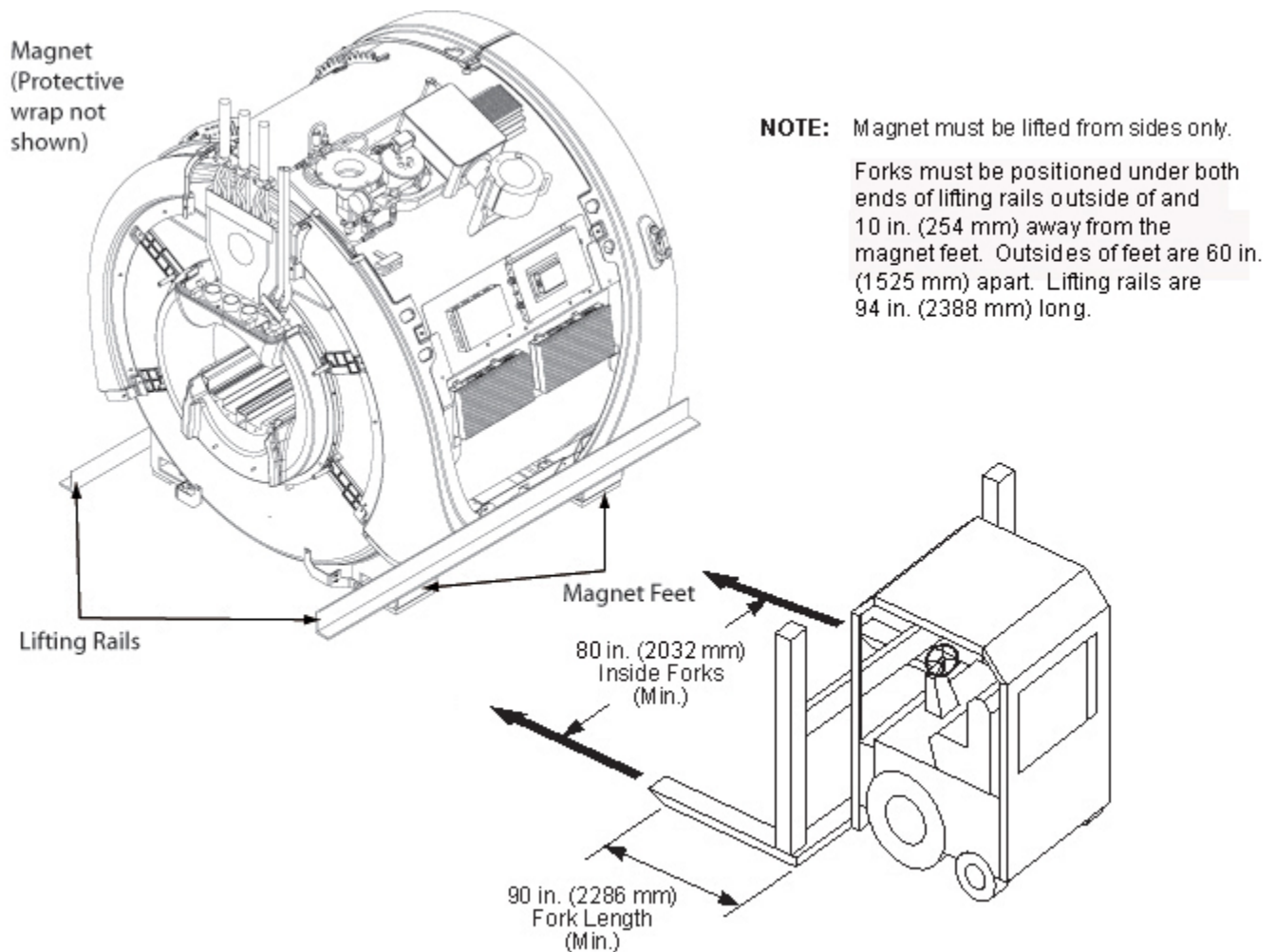


NOTICE

Extreme care must be exercised during forklift operation:

- Forklift must meet the minimum capacity and dimension requirements stated in [Table 5-1](#).
- The magnet must be picked up from the sides only. The forks must be placed under the lifting rails as indicated in [Illustration 5-4](#).
- The magnet must be lifted smoothly to avoid impact or jolts to the system, which may cause damage to the magnet.

Illustration 5-4: Forklift Under Lifting Rails

**NOTICE**

Forklift forks can damage the magnet enclosure or components. Use protective padding around the forks.

**NOTICE**

Impacts/jolts to the magnet while lifting/moving/lowering the magnet can cause expensive internal magnet damage. Lift/move/lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

Avoid tilting the magnet more than the maximum tilt as magnet damage may result.

1. Position a forklift that meets the specifications stated in [Table 5-1](#) at the side of the magnet facing the magnet. Locate the forks under the lifting rails to the outside of the magnet feet as indicated in [Illustration 5-4](#).

2. Wrap the full length of each fork with protective padding material to prevent damage to the magnet's enclosure.
3. Carefully drive the forklift until the forks are completely under both lifting rails in the areas shown in [Illustration 5-4](#).
4. Lift the forks to right below the lifting rails, adjust the distance between forks so that padded forks lightly touch the enclosure, and finish raising the forks to the lifting rails.
5. Lift the magnet with the forklift.
6. Smoothly move the magnet to the desired location and carefully lower to rest on a flat surface.

1.4.2 Unloading with a Crane



NOTICE

Do NOT crane lift a magnet that is on a pallet or inside a cage or crate. Crane lifting can only be done using the magnet lifting rails, which are not accessible while the magnet is inside a cage or crate. See [Chapter 6, International \(Air-Sea\) Magnet Delivery](#) for lifting a crated/palletted magnet.

[Section 1.4.2](#) is for crane lifting a magnet directly from a truck to nearby ground.

Consult [Section 1.4.3](#) before beginning any lifting operation from a truck to a raised opening in a building's exterior wall.



WARNING

POTENTIAL INJURY HAZARD.
THE MAGNET IS AN UNBALANCED LOAD. TILTING CAN RESULT IN A HAZARDOUS CONDITION THAT CAN RESULT IN MAGNET DAMAGE OR SERIOUS PERSONAL INJURY.

TO PREVENT TILTING:

- MAKE SURE THE LIFTING APPARATUS (CRANE, SPREADER BEAM, ETC.) MEETS THE SPECIFICATIONS STATED IN [TABLE 5-2](#) AND [ILLUSTRATION 5-1](#).
- ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.
- ADJUST LIFTING CABLES/SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF THE SURFACE.

**WARNING**

POTENTIAL INJURY HAZARDS.

TO PREVENT LIFTING CONDITIONS THAT MAY BE HAZARDOUS TO PERSONNEL AND DAMAGE THE MAGNET AND EQUIPMENT:

- MAKE SURE THE ENTIRE AREA WHERE LIFTING WILL OCCUR IS FREE OF OBSTRUCTIONS AND UNAUTHORIZED PERSONNEL.
- MAKE SURE THE SURFACE WHERE THE MAGNET WILL BE PLACED AFTER LIFTING IS FLAT.
- DO NOT CRANE LIFT A MAGNET DURING DANGEROUS WEATHER CONDITIONS.

**NOTICE**

Impacts/jolts to the magnet while lifting/moving/lowering the magnet can cause expensive internal magnet damage. Lift/move/lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

1. Make sure there are no obstructions in the area where lifting will occur and that a flat surface is available to position the magnet after lifting.
2. Position the hook of a crane and spreader beam, that meet the specifications stated in [Table 5-2](#), centrally over the magnet to ensure a vertical lifting force on the lifting cables/slings. (See [Illustration 5-1](#).)
3. Attach the rigging to the lifting cables/slings at each end of the magnet. (See [Illustration 5-1](#).)
4. When beginning to lift the magnet if the magnet does not remain close to level, carefully lower the magnet and adjust the lifting configuration, then lift the magnet again.
5. Smoothly move the magnet to the desired location, and carefully lower so it rests on a flat surface.

1.4.3 Crane Lift through Raised Opening in Exterior Wall

**WARNING**

POTENTIAL SERIOUS INJURIES.

SERIOUS INJURIES AND MAGNET/EQUIPMENT DAMAGE ARE POSSIBLE WHEN MOVING A MAGNET THROUGH A RAISED OPENING IN AN EXTERIOR WALL.

DO NOT BEGIN UNTIL:

- A COMPLETE WALK THROUGH OF THE CRANE LIFT PROCESS WAS PERFORMED PRIOR TO THE ACTUAL EVENT TO MAKE SURE ALL PROCESS DETAILS ARE COVERED.
- ALL NECESSARY EQUIPMENT IS ON SITE AND INSPECTED FOR SAFETY AND LOAD RATINGS.
- ALL NECESSARY PERSONNEL ARE TRAINED AND READY.

**WARNING**

POTENTIAL INJURY HAZARD.
MAGNET IS AN UNBALANCED LOAD. TILTING CAN RESULT IN A HAZARDOUS CONDITION THAT CAN RESULT IN MAGNET DAMAGE OR SERIOUS PERSONAL INJURY.

TO PREVENT TILTING:

- MAKE SURE THE LIFTING APPARATUS (CRANE, SPREADER BEAM, ETC.) MEETS THE SPECIFICATIONS STATED IN [TABLE 5-2](#) AND [ILLUSTRATION 5-1](#).
- ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.
- ADJUST LIFTING CABLES/SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF SURFACE.

**WARNING**

POTENTIAL INJURY HAZARDS

TO PREVENT LIFTING CONDITIONS THAT MAY BE HAZARDOUS TO PERSONNEL AND DAMAGING TO THE MAGNET AND EQUIPMENT:

- MAKE SURE THE ENTIRE AREA WHERE LIFTING WILL OCCUR IS FREE OF OBSTRUCTIONS AND UNAUTHORIZED PERSONNEL.
- MAKE SURE THE SURFACE WHERE THE MAGNET WILL BE PLACED AFTER LIFTING IS FLAT.
- DO NOT CRANE LIFT A MAGNET DURING DANGEROUS WEATHER CONDITIONS.

**NOTICE**

The lifting hooks of the spreader beam **MUST** be located as shown in . The magnet can be lifted by a crane that meets the specifications listed in [Table 5-2](#). Failure to follow these requirements can result in serious damage to the magnet.

1. Verify that the opening is **at least** 96 in. (2,440 mm) wide and 102 in. (2,590 mm) tall, minimum. The magnet should pass through the opening side first without hitting the opening. (A larger opening will make both the operation easier and accidental magnet damage less likely.)
2. Position the hook of a crane and spreader beam that meet the specifications stated in [Table 5-2](#) centrally over the magnet to ensure a vertical lifting force on the lifting cables/slugs. (See [Illustration 5-1](#).)
3. Rig the magnet with chain hoists and spreader beam toward the building, lifting cables/slugs away from the building. (See [Illustration 5-1](#).)
4. Attach lifting straps to both ends of the lifting rail to face toward the building. (See [Illustration 5-1](#).)

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Chapter 6 International (Air-Sea) Magnet Delivery

1 International (Air-Sea) Magnet Delivery



WARNING

POTENTIAL ASPHYXIATION HAZARD.
LOSS OF MAGNET VACUUM WILL RESULT IN THE RAPID EXPULSION OF HELIUM GAS, WHICH CAN CAUSE ASPHYXIATION IN ENCLOSED AREAS.

USE EXTREME CAUTION TO NOT CONTACT OR DAMAGE THE VACUUM VESSEL DURING MAGNET TRANSIT OR SITING.

1.1 Equipment and Tools



NOTICE

Do NOT use a crane to lift a magnet that is on a pallet or inside a crate. Crane lifting can only be done using the magnet lifting rails, which are not accessible while the magnet is inside the crate.

Table 6-1: Crated Magnet Lifting/Handling Information

Item	Equipment Specification/Rating		Furnished By	Intended Use
Forklift	Forklift Capacity:	27,250 lbs. (12,360 kg) at 50 in. (1,270 mm) to load center, minimum	Rigger	Unloading or moving magnet, pallet and crate together.
	Minimum Fork Length:	93 in. (2,362 mm)		
	Minimum Distance Between Forks:	Old style pallet distance is 40 in. (1,016 mm) New style pallet (used after November 2009) distance is 81.1 in. (2,060 mm)		
Crane (On-Site Only)	Total Load:	27,250 lbs. (12,360 kg), minimum		Crate removal and unloading or moving magnet (no pallet or crate)
Two-Legged Cable Bridle/Sling (Lifting Rail to Spreader Beam)	Total Load:	27,250 lbs. (12,360 kg), minimum		
	Leg Length:	90 in. (2,286 mm) minimum, all legs same length		
	Load per Leg:	13,625 lbs. (6,180 kg), minimum		
Two-Legged Cable Bridle/Sling (Spreader Beam to Crane)	Leg Length:	90 in. (2,286 mm) minimum, all legs same length		
	Load per Leg:	27,250 lbs. (12,360 kg), minimum		
Sling (Strap or Cable)	Minimum Length:	72 in. (1,830 mm)		Crate removal.
	Work Load Limit:	1 ton (907 kg)		
Shackle or Hook	Work Load Limit:	1 ton (907 kg)		
Wrench	Sizes:	9/16 and 3/8 in.		

Note: For handling and lifting magnet outside the crate, refer to [Chapter 5, Domestic Magnet Delivery](#).

1.2 In-Transit Service



NOTICE

In-transit service **MUST** be performed by qualified personnel only and in **STRICT** conformance with the Liquid Helium Fill procedure stated in the appropriate manual:

- Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem*
- Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem*
- *2301164PRE MR Magnet Safety Requirements*, which is included with both of the above manuals.

These documents are available through the Common Documentation Library at gehealthcare.com or your GE Healthcare Field Service Representative.

In-transit helium refill is performed based upon the magnet shipping date per the documents stated above. Access the plumbing through the access panel identified in [Illustration 6-1](#). In-transit plumbing is configured as shown in [Illustration 6-2](#).

In-transit electrical checks are covered in [Chapter 7, Magnet System Checks \(by GE Service\)](#).

Illustration 6-1: Crate/Pallet

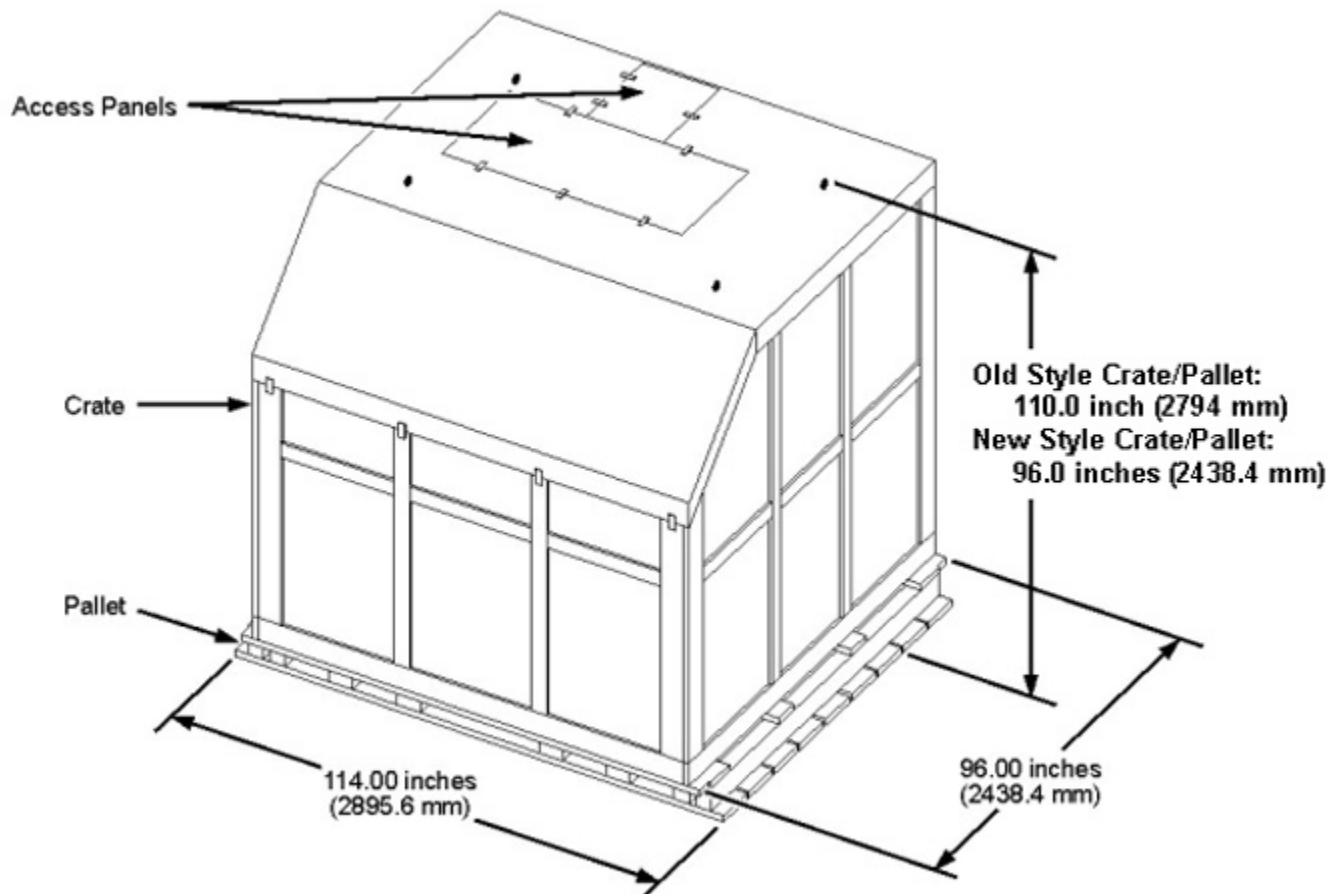
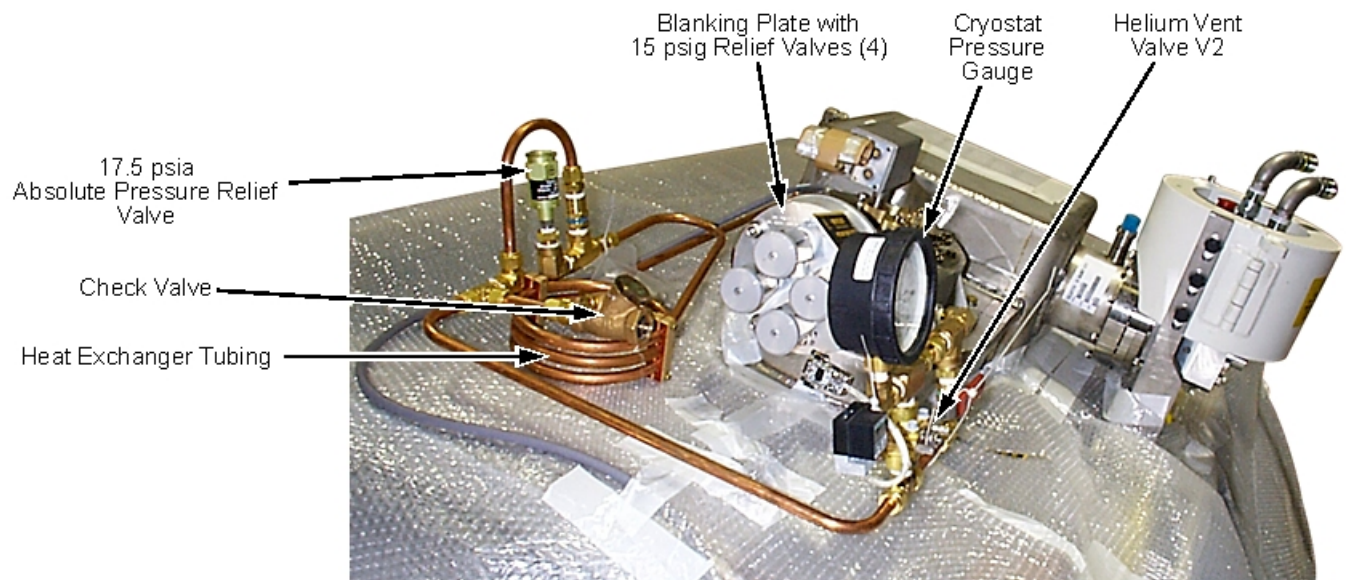


Illustration 6-2: Magnet Plumbing Inside Top Service Access Panel



1.3 In-Transit Handling

1.3.1 In-Transit Handling with Crane



NOTICE

Do NOT lift the magnet/crate/pallet package using a crane. Crane lifting can only be done using the magnet lifting rails, which are not accessible while the magnet is inside the crate.

1.3.2 In-Transit Handling with Forklift

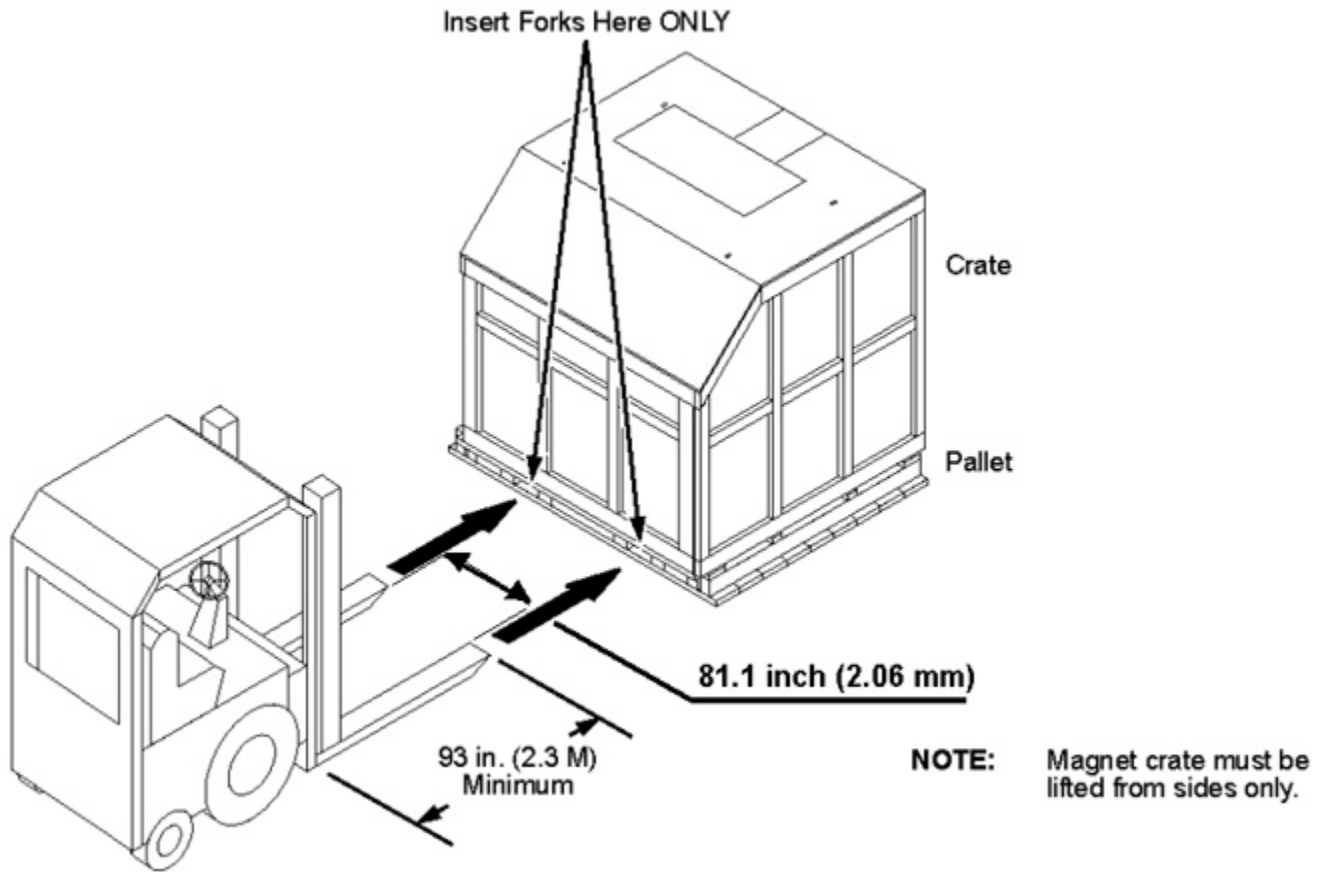


NOTICE

Extreme care must be exercised during forklift operations:

- Forklift must meet the minimum capacity and dimension requirements stated in [Table 6-1](#).
- The magnet must be picked up from the sides only and only where shown in [Illustration 6-3](#).
- The magnet must be lifted smoothly to avoid impact or jolts to the system that may cause damage to the magnet.

Illustration 6-3: Forklift Lifting Points of Magnet/Crate/Pallet Package

**NOTICE**

Impacts/jolts to the magnet while lifting/moving/lowering the magnet/crate/pallet package can cause expensive internal magnet damage. Lift/move/lower smoothly. Do not allow the crate/pallet to bump or hit anything forcefully.

Avoid tilting the magnet/crate/pallet package more than the maximum tilt specified in as magnet damage may result.

1. Position a forklift beside the magnet crate/pallet meeting the requirements stated in [Table 6-1](#) and shown in [Illustration 6-3](#).
2. Carefully insert the forklift forks completely into the holes provided in the long side of the shipping pallet. (See [Illustration 6-3](#).)
3. Smoothly lift the crate/pallet with the forklift, and move the crate/pallet to the desired location.
4. Lower the entire package to rest on a flat, smooth surface. Do **not** rest magnet/crate/pallet on any surface that is not flat or strong enough to support the magnet/crate/pallet package.

1.4 On Site

1.4.1 Crate Removal



NOTICE

Do NOT lift the magnet/crate/pallet package using a crane. Crane lifting can only be done using the lifting rails, which are not accessible while the magnet is inside the crate.

Remove the magnet's shipping crate while the magnet/crate/pallet package is on the trailer/transport.



NOTICE

Damage to the magnet and/or magnet enclosures may result if the crate is removed while the crate's side panels are in their closed (shipping) position, or if the crate is disassembled while the magnet is inside the crate.

1.4.1.1 Removal of Subsystem Crates



NOTICE

During this procedure, carefully inspect all packaging and contents for damage that may have occurred during shipping.

1. Remove all subsystem crates, except the magnet crate, from the trailer/transport using a crane or forklift. Inspect all crates for visible damage. Use the procedure in Damage In Transportation to report any damage found.
2. Move subsystem crates to a receiving location protected from the weather, preferably close to and at the same level as the MR Suite/Magnet Room.

1.4.1.2 Positioning Crane

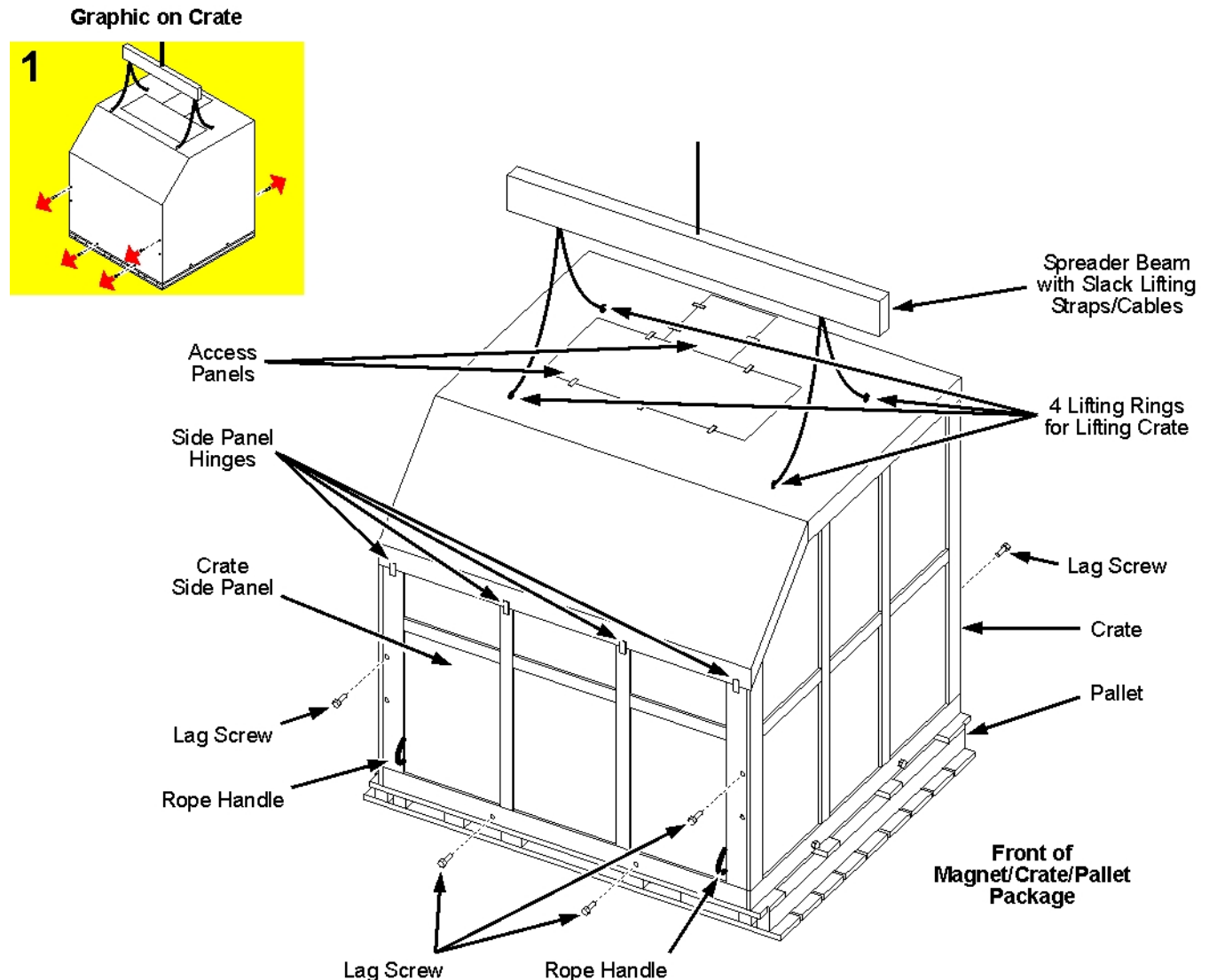


NOTICE

Do not remove any lag screws on the crate or pallet until the crane is positioned and the slings/cable bridles are attached to the lifting rings on the crate.

1. Position a crane above the center of the crate meeting the specifications stated in [Table 6-1](#). Either a hook/shackle and slings or a spreader beam and slings/cable bridles meeting the specifications stated in [Table 6-1](#) may be used during crate removal.
2. Attach the slings/cable bridles to the lifting rings on top of the crate. (See [Illustration 6-4](#).)

Illustration 6-4: Lag Screw Removal to Open Crate Side Panels



1.4.1.3 Shipping Crate Lifting Configuration

1. Remove the lag screws along the lower edges of the crate's left and right side panels that secure those panels to the pallet. (See [Illustration 6-4](#) and graphic 1 on the shipping crate.)
NOTE: Do **not** remove the lag screws securing the crate's front and back end panels to the pallet until **after** the crate's side panels are secured in their open position.
2. Remove the lag screws securing the crate side panels to the crate end panels. (See [Illustration 6-4](#) and graphic 1 on the shipping crate.)
NOTE: Do **not** remove the hinges along each crate side panel's upper edge.
3. Pull open the crate's left and right side panels using the rope handle on each side panel. (See [Illustration 6-5](#) and graphic 2 on the shipping crate.)
NOTE: The crate's left and right side panels are hinged along their upper edges.

4. Swing the wooden block found inside each vertical corner of the crate outward. (See [Illustration 6-6.](#))
5. Attach the loose end of each block to the adjoining side panel using one of the lag screws removed previously. (See [Illustration 6-6.](#))
6. Remove the lag screws along the lower edges of the crate's front and back end panels that secure those panels to the pallet. (See [Illustration 6-5](#) and graphic 2 on the shipping crate.)

Illustration 6-5: Securing Crate Side Panels in Open Position

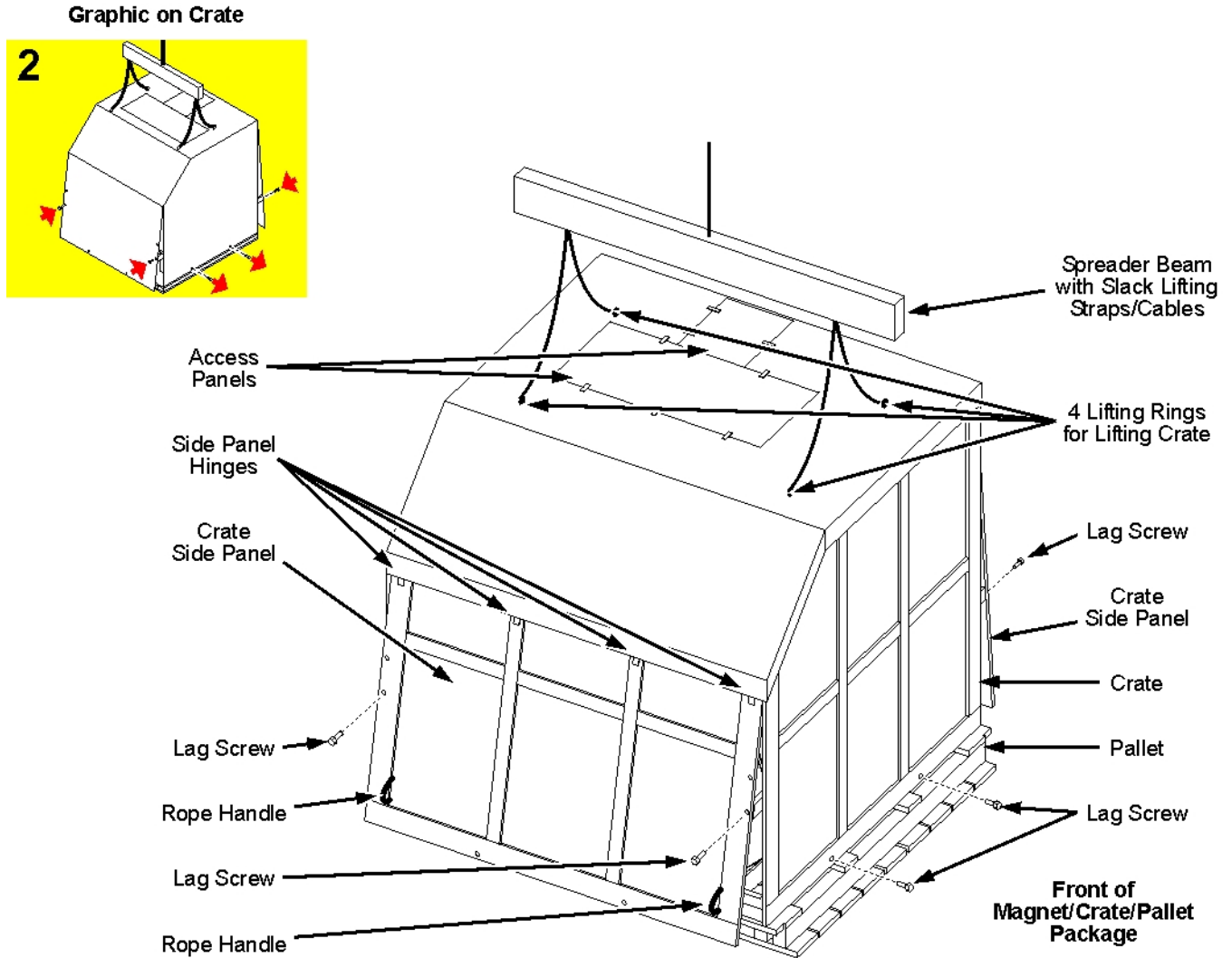
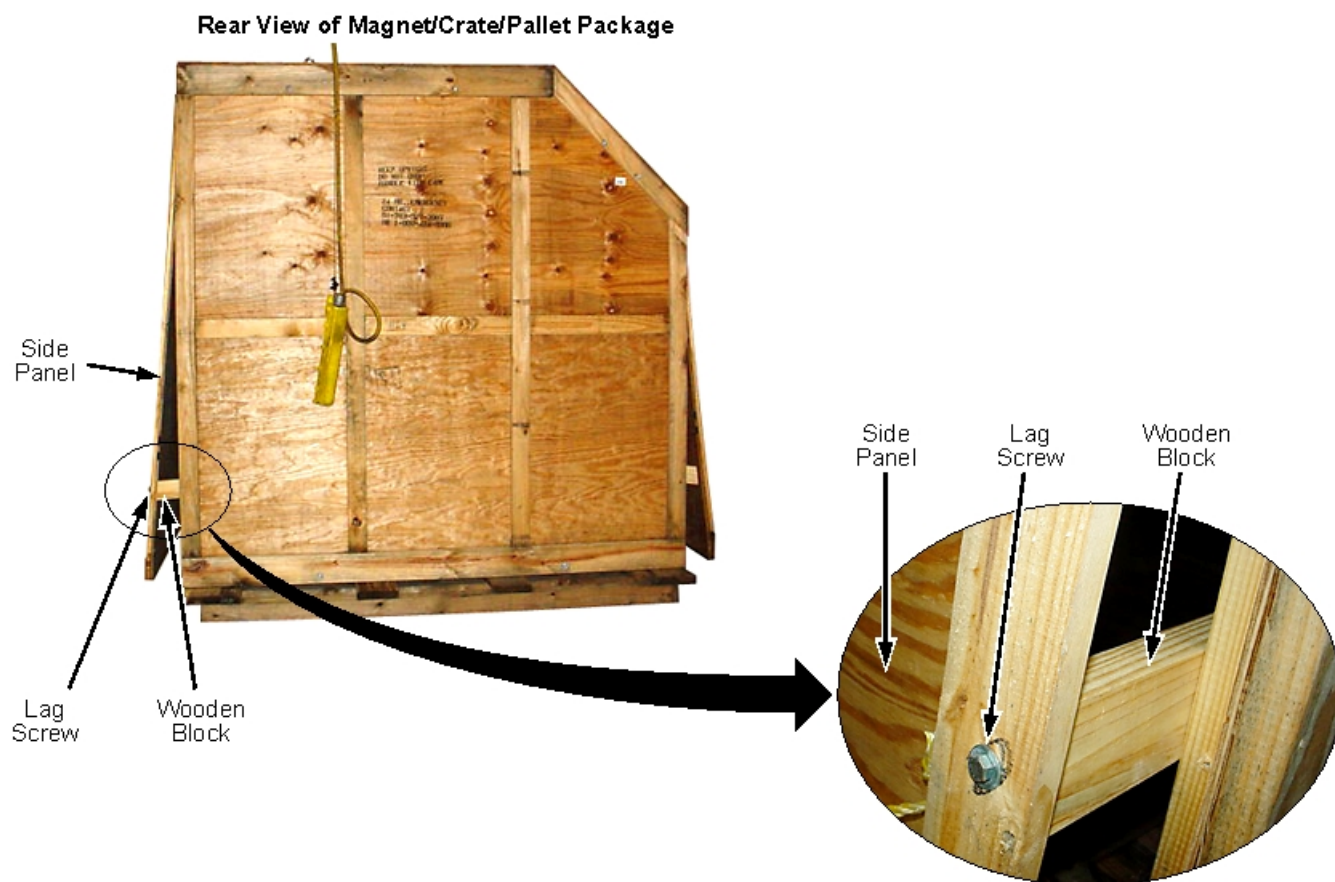


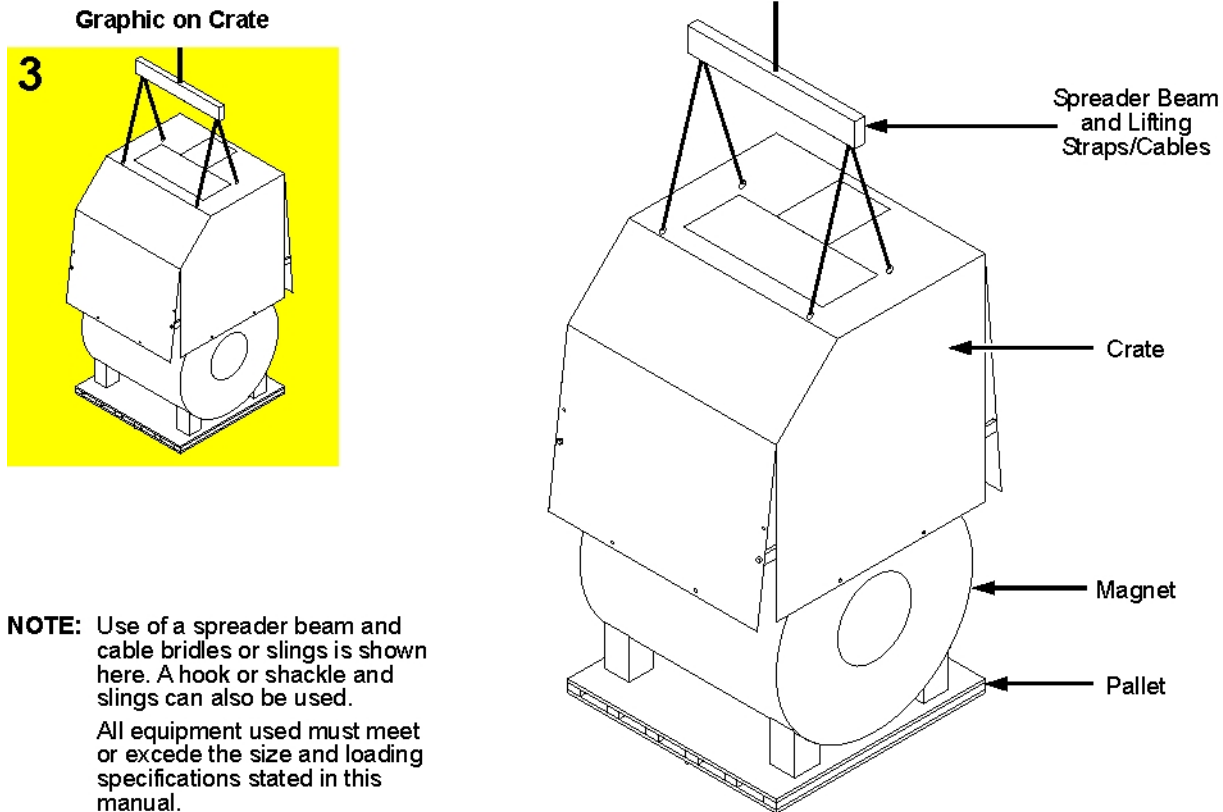
Illustration 6-6: Air/Sea Shipping Crate - Side Panels Raised



1.4.1.4 Lifting Crate Off Pallet

1. Tighten the slings/cable bridles and carefully begin lifting the crate. (See [Illustration 6-7](#) and graphic 3 on the shipping crate.) If the crate does not remain approximately level, carefully lower the crate and adjust the lifting configuration, then lift the crate again.
2. Crane lift the crate off the magnet's shipping pallet and fully above the magnet; place the crate in a clear area.

Illustration 6-7: Crate Lifting Using Crane



1.4.2 Removing Magnet from Pallet



WARNING

POTENTIAL INJURY HAZARD
PUSHING MAGNET ENCLOSURES MAY RESULT IN BODILY INJURY TO PERSONNEL.

DO NOT PUSH MAGNET ENCLOSURES.



NOTICE

To prevent magnet damage:

- Do not apply any force to the magnet enclosures. (See [Illustration 6-8.](#))
- Only use equipment/tools that meet the specifications stated in [Table 6-1](#) to crane lift or forklift the magnet off its shipping pallet.
- Before moving magnet to MR Suite, refer to [Chapter 9, Moving Magnet to MR Suite.](#)

Illustration 6-8: Do Not Push Signs



One on Left Side
One on Right Side



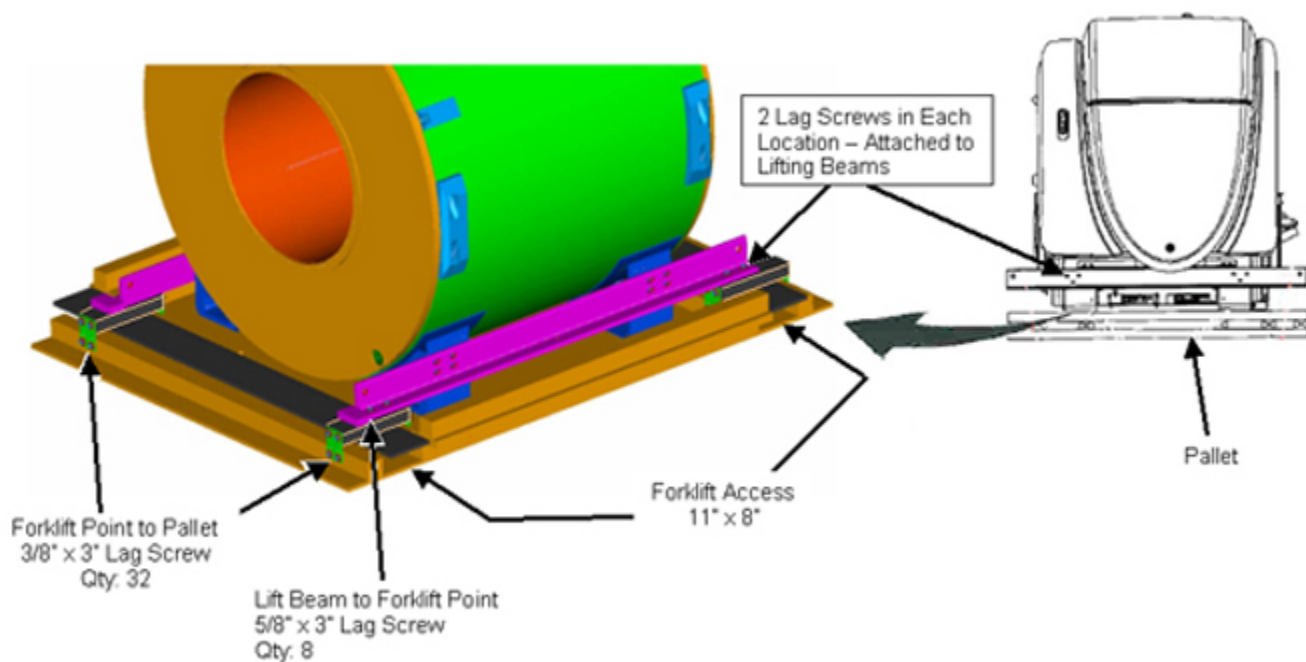
Two on Front End
Two on Rear End

**NOTICE**

Do not move the magnet to the MR Suite while the magnet is on its shipping pallet.

1. Unbolt and remove the four 1 in. (25.4 mm) nuts securing the magnet to the pallet through the ORANGE lifting beams. (See [Illustration 6-9](#).)
2. Rig and lift the magnet off the pallet in conformance with the Forklift Unloading (when using a forklift) or Crane Unloading (when using a crane) subsections of [Chapter 5, Domestic Magnetic Delivery](#).

Illustration 6-9: Unbolting Magnet from Shipping Pallet



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Chapter 7 Magnet System Checks (by GE Service)

1 Magnet System Checks (by GE Service)

1.1 Physical Inspection

1. Locate the *Pre-Delivery Information Package* shipped with the magnet. It contains the Bill of Material for the delivered magnet system. Check that all boxes indicated are present.
2. Check the contents of each box against its packing list when the boxes are brought into the MR site.
3. Inspect the magnet for physical damage and icing/condensation on the body. If no problem is found, unload the magnet.

NOTE: Because of the higher boil-off and helium gas flow through the Vertical Penetration on international shipments, some frost on the Vertical Penetration may be normal during periods when the Coldhead has been shut off.

4. If icing or condensation are present on the exterior or the bore of the magnet, check the liquid helium level before unloading. The occurrence of icing needs to be understood before filling the magnet with helium. Refer to the *Magnet Monitor Installation* subsection in the Set-Up and Calibration chapter in the appropriate manual:
 - For HDi and HDx: Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem*
 - For MR750: Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem*

NOTE: If the magnet has been sitting for a period of time with the Coldhead inoperative, the magnet may be depleted of cryogenics. If so, the magnet can be filled when magnet gets bolted down in the magnet room.

1.2 Electrical Checks



NOTICE

Magnet Electrical Checks must be done after the magnet is sited in the MR Suite.



NOTICE

All superconducting coils measured must be cold (4.2K) for the resistance values shown in [Table 7-1](#), [Table 7-2](#) and [Table 7-3](#) to be accurate.

1. Make sure the Shim Lead is engaged. If needed, re-engage the Shim Lead in conformance with the *Shim Lead Engagement and Disengagement* section in the appropriate manual: Set-Up and Calibration chapter in the appropriate manual:
 - For HDi and HDx: Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem*
 - For MR750: Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem*
2. Locate the connector pins using [Table 7-1](#), [Table 7-2](#), [Table 7-3](#) and [Illustration 7-1](#).

NOTE: See the *Magnet System Wiring* subsection in the Schematics & Interconnects chapter for wiring diagrams in the appropriate manual:

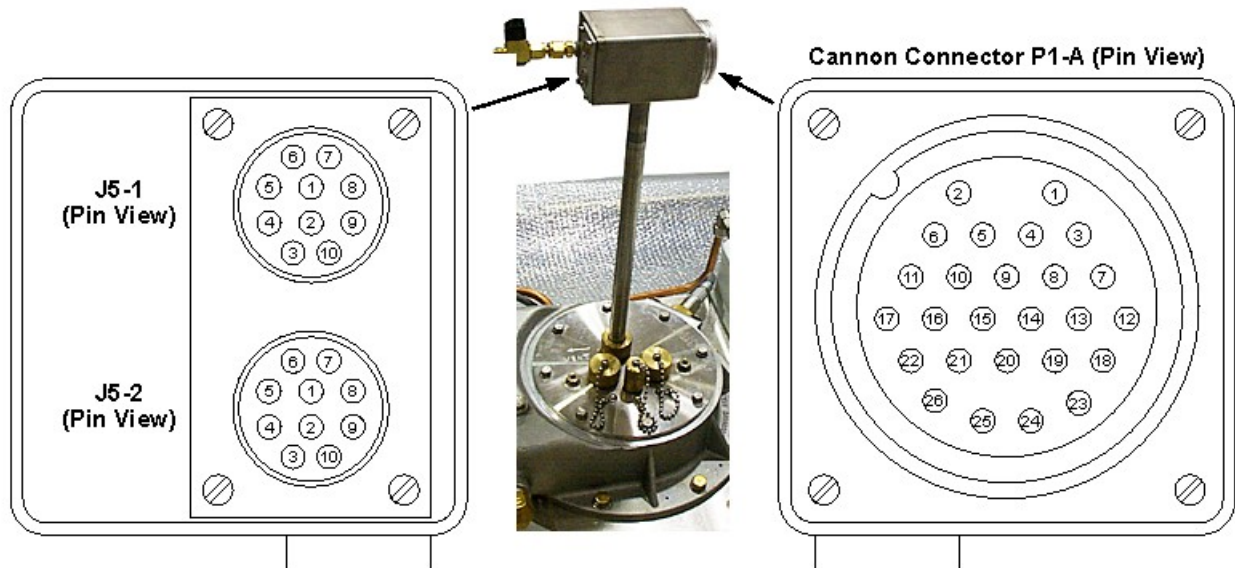
- For HDi and HDx: Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem*
- For MR750: Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem*

3. Use a digital meter to measure the resistance across the identified connector pins.
4. Measure the resistances and record, under “Measured”, Main Coil resistances in [Table 7-1](#), S/C Shim Coil resistances measured on the Shim Lead in [Table 7-2](#), and S/C Shim Coil resistances measured on the Shim Lead Pigtail Cable in [Table 7-3](#). Compare measured resistances to the “Typical” values listed.
5. Perform the checks identified in [Table 7-4](#). If any problems are found, the Shim Lead can be retracted to isolate shorts from the Shim Coils.
6. Report any damage found in conformance with the Damage In Transportation in the front of this manual.
7. Report all problems found to the regional Magnet & Cryogenics (MAC) Team Leader.

NOTE: During normal operation, the Shim Lead Assembly will be in the disengaged position and rotated counterclockwise until assembly locks into position.

Illustration 7-1: Connector Pinouts

Shim Lead (Main Extension Lead) Housing



Shim Lead Pigtail Cable (2135362)

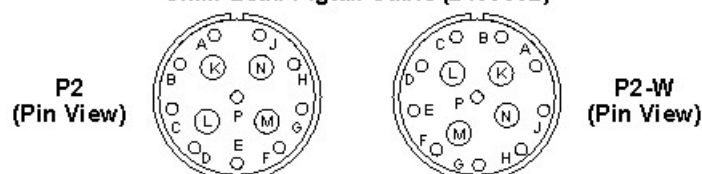


Table 7-1: Main Coil Resistance Checks - Cold (4.2K)

Connector	Pins+, -	Resistance (Ohms)	
		TypicalWB Series	Measured
Main Coil Power Lugs	+, -	<6	
J5-1 on Shim Lead	9, 10		

Table 7-2: Superconducting Shim Coil Resistance Checks - Cold (4.2K), Measured on Shim Lead

Function		Shim Lead Con- nector	Pins+, -	Resistance (Ohms)	
Shim PS Setting	Shim Coil Circuit			TypicalWB Series	Measured
AX1	z^1	P1-A	1, 19	0.2 - 0.8	
AX2	z^2		2, 20		
AX3	z^3		3, 21		
AX4	z^4		4, 22		
AX5	z^5		5, 23		
AX6	z^6		6, 24		
T1-1	$z(x^2-y^2)$		13, 23		
T1-2	X		16, 19		
T1-3	z^2x		14, 21		
T1-4	ZX		17, 20		
T1-5	x^2-y^2		15, 22		
T1-6	x^3		18, 24		
T2-1	ZXY		11, 23		
T2-2	Y		9, 19		
T2-3	z^2y		7, 21		
T2-4	ZY		10, 20		
T2-5	XY		8, 22		
T2-6	y^3		12, 24		
S/C Switch Heaters Main Switch		J5-1 or J5-2	1, 2	22 - 27	
Axial Shims			5, 6	18 - 22	
Transverse 1			7, 6	13 - 16	
Transverse 2			8, 6		

Table 7-3: Superconducting Shim Coil Resistance Checks - Cold (4.2K), Measured on Shim Lead Pigtail Cable

Function		Shim Lead Pigtail Cable Connector	Pins+, -	Resistance (Ohms)	
Shim PS Setting	Shim Coil Circuit			Typical WB Series	Measured
AX1	Z ¹	P2	A, (K,L)	0.2 - 0.8	
AX2	Z ²	P2-W			
AX3	Z ³	P2	B, (M,N)		
AX4	Z ⁴	P2-W			
AX5	Z ⁵	P2	C, P		
AX6	Z ⁶	P2-W			
T1-1	Z(X ² -Y ²)	P2	D, P		
T1-2	X		F, (K,L)		
T1-3	Z ² X		G, (M,N)		
T1-4	ZX	P2-W	F, (K,L)		
T1-5	X ² -Y ²		G, (M,N)		
T1-6	X ³		D, P		
T2-1	ZXY	P2	E, P		
T2-2	Y		H, (K,L)		
T2-3	Z ² Y		J, (M,N)		
T2-4	ZY	P2-W	H, (K,L)		
T2-5	XY		J, (M,N)		
T2-6	Y ³		E, P		

Table 7-4: Shim Lead Checks

	P1-A Connector Pins on Shim Lead		Pigtail Connectors ^a			
Type of Check	From	To	From	To	Acceptable Range	Measured
Checks for Lead-to- Lead Shorts	19	20	P2 (K,L)	P2-W (K,L)	>16K Ohms	
		21		P2 (M,N)		
		22		P2-W (M,N)		
		23		P2 (P)		
		24		P2-W (P)		
	20	21	P2-W (K,L)	P2 (M,N)		
		22		P2-W (M,N)		
		23		P2 (P)		
		24		P2-W (P)		
	21	22	P2 (M,N)	P2-W (M,N)		
		23		P2 (P)		
		24		P2-W (P)		
	22	23	P2-W (M,N)	P2 (P)		
		24		P2-W (P)		
	23	24	P2 (P)	P2-W (P)		
Checks for Lead-to-Ground Shorts	19	Ground ^b	P2 (K,L)	Ground		
	20		P2-W (K,L)			
	21		P2 (M,N)			
	22		P2-W (M,N)			
	23		P2 (P)			
	24		P2-W (P)			

^a Paired pins inside parenthesis are interconnected inside the pigtail connector: P2 (K,L) to P2-W (K,L) means Connector P2, pins K and/or L, to Connector P2-W, pins K and/or L.

^b Ground defined as Shim Lead Cover.

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Chapter 8 VibroAcoustic Damping Mat Placement

1 Surface Mounted VibroAcoustic Damping Mat Placement

The VibroAcoustic Damping Mats **must** be installed before moving the magnet into the magnet room.



CAUTION

Heavy Object

Each VibroAcoustic Damping Mat weighs ~250 lbs. (~105 kg). Lifting mats alone without assistance can result in personal injury.

Do not lift/move mats without mechanical assistance or at least four people.



NOTICE

The Surface Mounted VibroAcoustic Damping Mats must be installed on the floor or on the magnet before moving the magnet into the magnet room.



NOTICE

Tasks in this section are to be performed by riggers, not by GE Service Personnel

NOTE: The VibroAcoustic Damping Mat may arrive in a cold state preventing compression. The VibroAcoustic Damping Mat should be placed in the magnet room prior to moving magnet to MR suite to allow for temperature stabilization. If the magnet exceeds the installation height specification, allow the magnet to settle on the VibroAcoustic Damping Mat for 24 hours, then remeasure the magnet height.

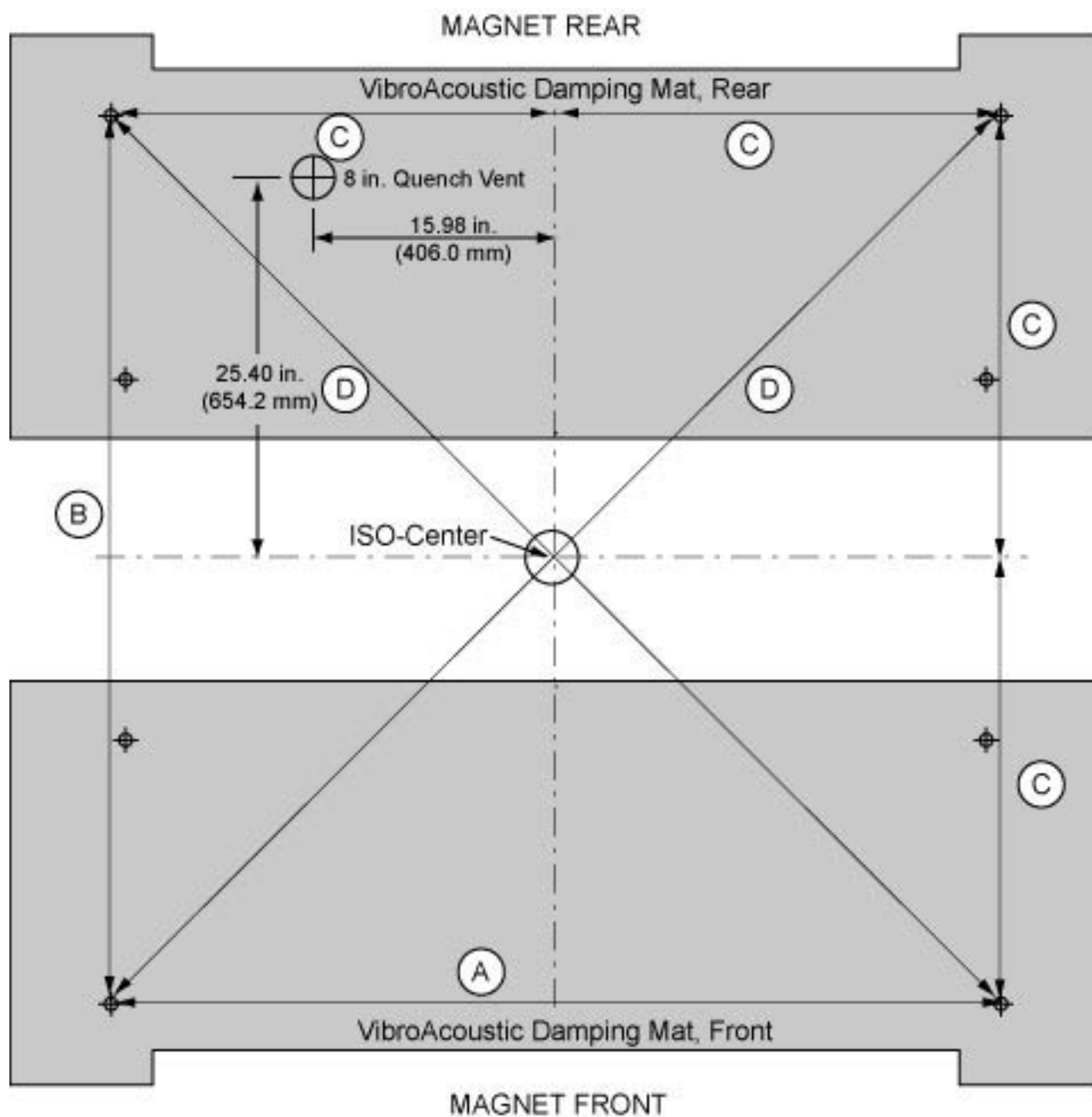
Table 8-1: VibroAcoustic Damping

Quantity	Description
2	VibroAcoustic Damping Mat, one labeled FRONT and one REAR
4	3 in. x 3 in. (76.8 mm x 76.8 mm) Aluminum Washers*
4	0.75-10 x 4 in. Stainless Steel Studs*
4	0.75-10 Stainless Steel Nuts*
4	Stainless Steel Flat Washers*

* For anchoring the magnet to VibroAcoustic Damping Mats, refer to [Chapter 10, Magnet Leveling, Foot Shimming and Bolt Down](#).

1. Verify that the VibroAcoustic Damping Mat Kit contains the parts listed in [Table 8-1](#).
2. Remove any debris from the magnet room floor where the Mats will be positioned.
3. Verify that there are no RF seams under the VibroAcoustic Damping Mats.
4. The VibroAcoustic Damping Mats consist of two identical plates. Place VibroAcoustic Damping Mats as shown in [Illustration 8-1](#).

Illustration 8-1: Positioning VibroAcoustic Damping Mats

**Dimensions**

- A** 53 inch (1356.8 mm) Left-Right
- B** 53 inch (1356.8 mm) Front-Back
- C** 26.5 inch (673.1 mm)
- D** 74.95 inch (1903.8 mm) Diagonal

NOTE: [Illustration 8-2](#) shows the surface-mounted VibroAcoustic Damping Mats.

Illustration 8-2: Surface-Mounted VibroAcoustic Damping Mats



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Chapter 9 Moving Magnet to MR Suite

1 Moving Magnet to MR Suite



WARNING

POTENTIAL ASPHYXIATION HAZARD
LOSS OF MAGNET VACUUM WILL RESULT IN THE RAPID EXPULSION OF HELIUM GAS, WHICH CAN CAUSE ASPHYXIATION IN ENCLOSED AREAS. MAKE SURE HOSPITAL PERSONNEL ARE AWARE OF THIS SITUATION PRIOR TO MAGNET BEING MOVED WITHIN THE ENCLOSED AREAS.

USE EXTREME CAUTION TO AVOID CONTACT OR DAMAGE TO THE VACUUM VESSEL DURING MAGNET TRANSIT OR SITING.



WARNING

POTENTIAL INJURY HAZARD
PUSHING MAGNET ENCLOSURES MAY RESULT IN BODILY INJURY TO PERSONNEL.

DO NOT PUSH MAGNET ENCLOSURES.



NOTICE

To prevent magnet damage do NOT apply any force to the magnet's enclosures.



NOTICE

Tasks in this section are to be performed by Riggers, NOT by GE Service Personnel

Illustration 9-1: Do Not Push Signs



One on Left Side
One on Right Side



Two on Front End
Two on Rear End

**WARNING**

POTENTIAL INJURY HAZARD
MOVING THE MAGNET IMPROPERLY WILL CAUSE PERSONAL INJURY OR
MAGNET DAMAGE.

REFER TO
[CHAPTER 5, DOMESTIC MAGNET DELIVERY](#)

BEFORE MOVING THE MAGNET USING A FORKLIFT OR CRANE.

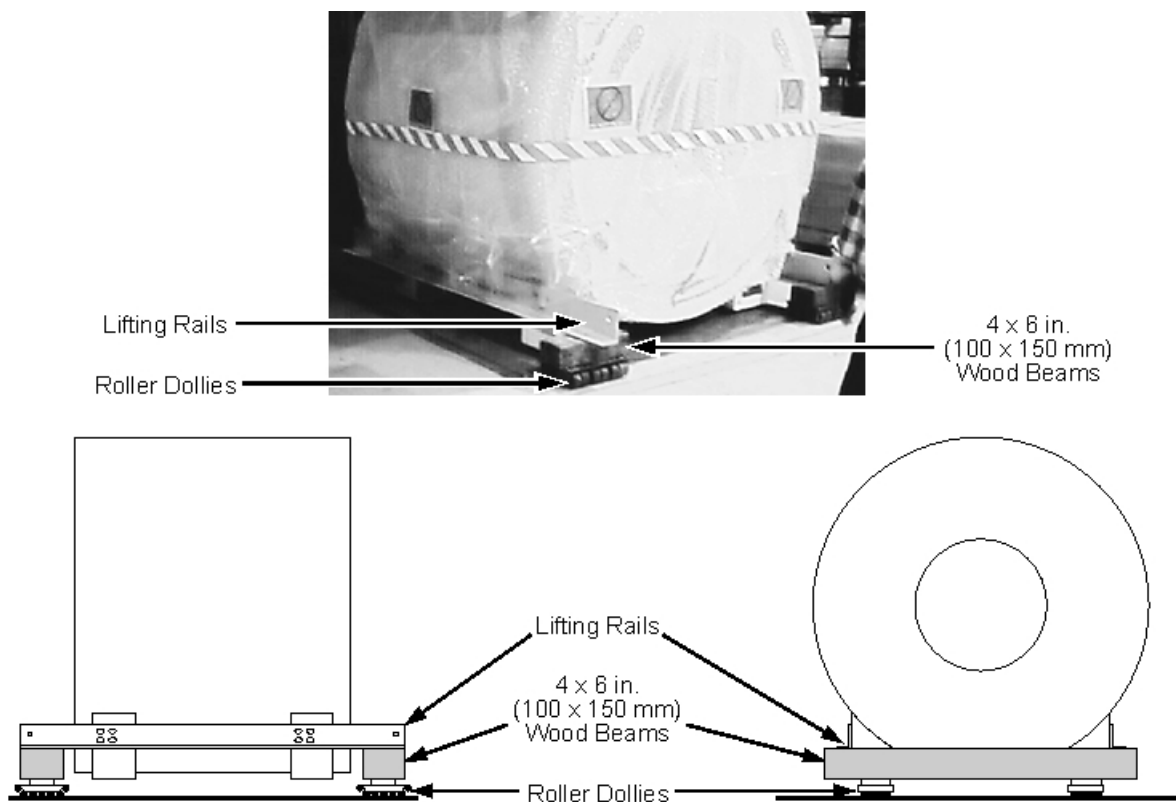
**NOTICE**

Magnet and dock installation and placement are critical to image quality and hardware reliability. Magnet must be installed level with the isocenter to the specifications listed in Magnet Leveling, Foot Shimming and Bolt

1.1 Introduction

Once the magnet is moved to the building using a crane or forklift (covered in [Chapter 5, Domestic Magnet Delivery](#)), it needs to be moved to the Magnet Room. Roller dollies in the arrangement (shown below) are recommended for moving the magnet inside a building. Place steel floor plates along the magnet delivery route when using roller dollies.

Illustration 9-2: Magnet on Roller Dollies



There are many methods to help move the magnet, including the use of a motorized tow vehicle, a come-along, or a chain jack. Any cables, chains, or straps used must be attached to the lifting rails.

**NOTICE**

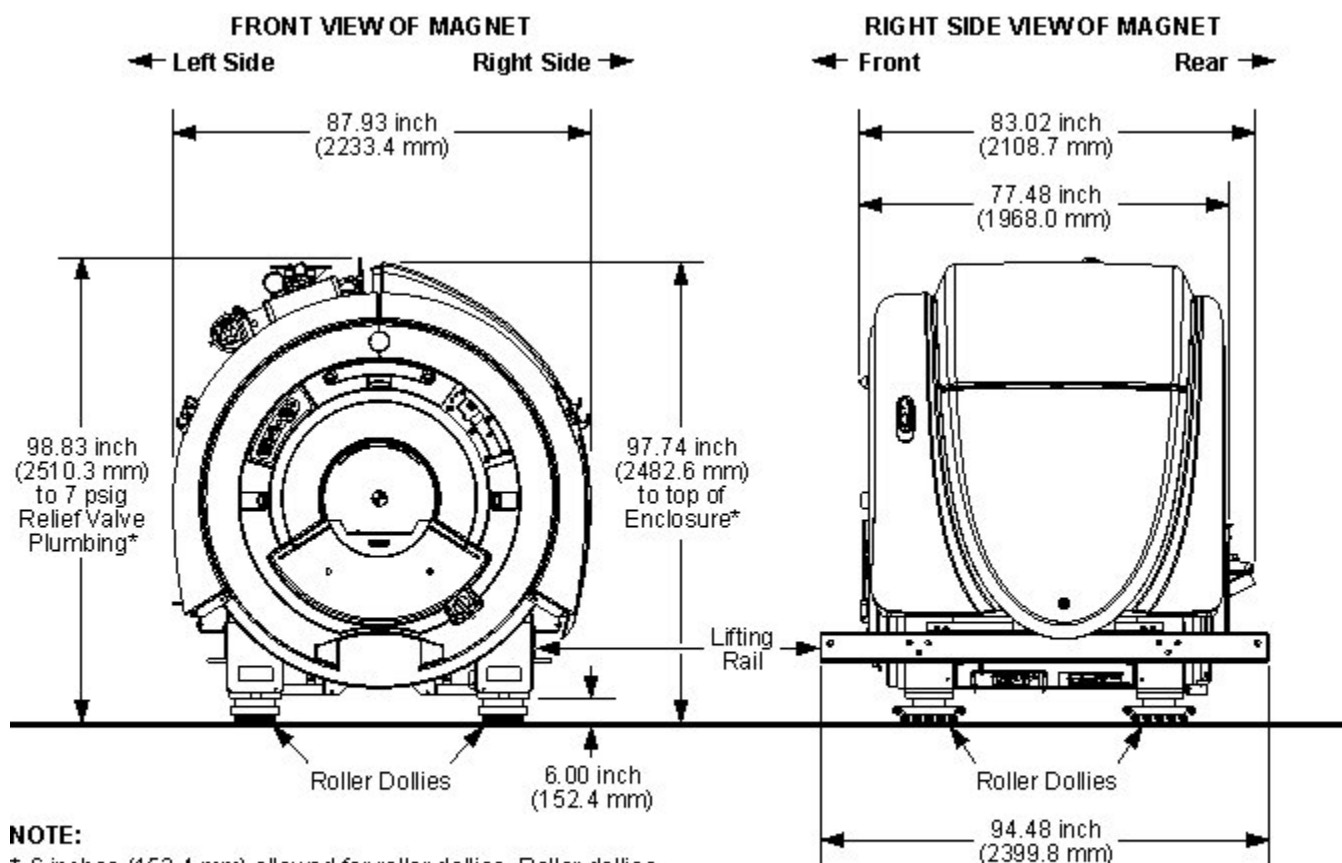
Any floor anchors that are used must NOT penetrate the RF Shield.

1.2 Moving the Magnet

1.2.1 Preparation

1. Verify magnet front-rear orientation is relative to the Magnet Room's front and rear.
2. Check all clearances along the route the magnet will move to the Magnet Room. Compare those clearances with the appropriate illustration below.

Illustration 9-3: Clearance Dimensions - LCC Magnet with Roller Dollies Attached

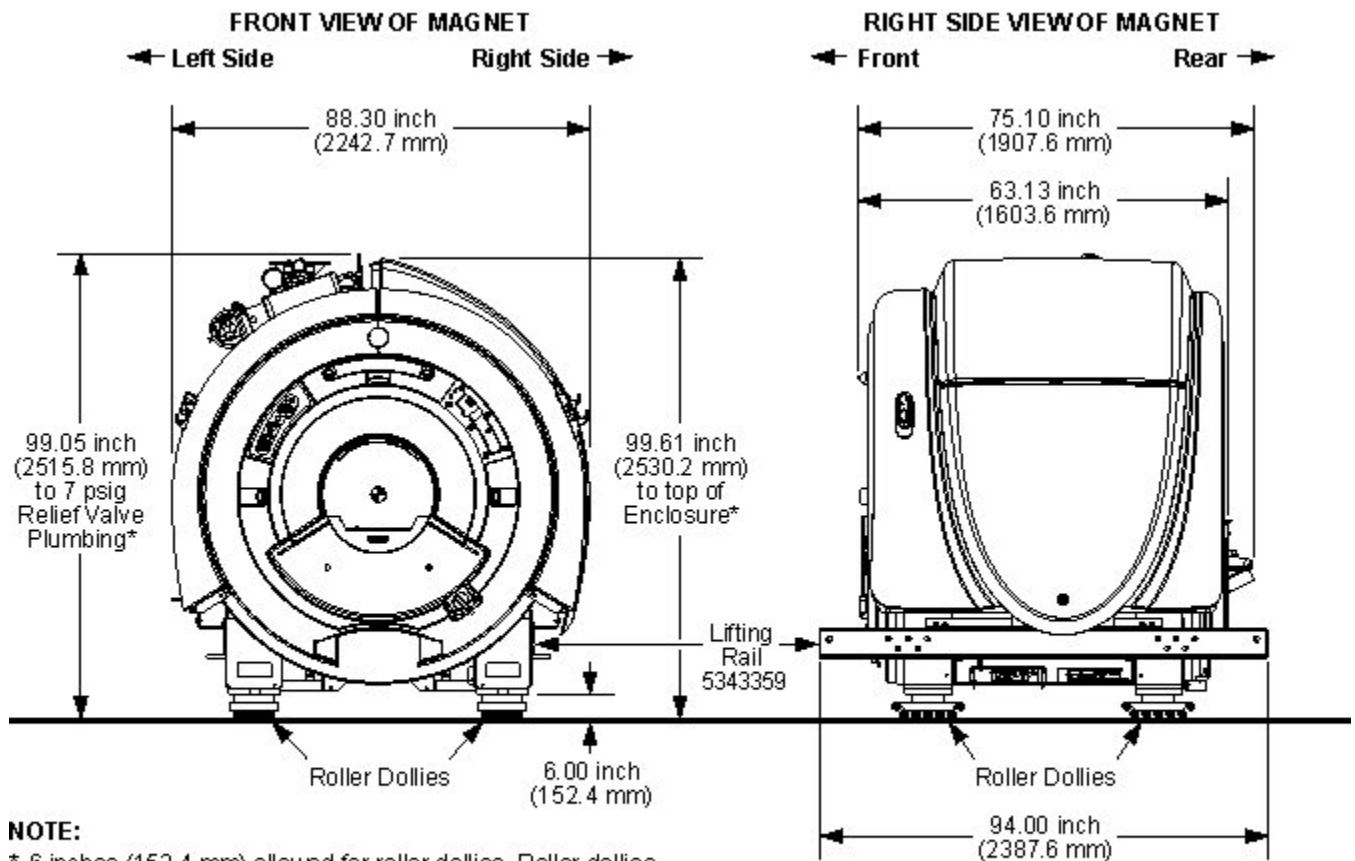
**NOTE:**

* 6 inches (152.4 mm) allowed for roller dollies. Roller dollies taller than 6 inches (152.4 mm) increase overall height.

Minimum height for service clearance is 105 inches (2667 mm). This clearance is needed for Main Lead Extensions, Shim Lead and fill line installation.

Source: 5162096IDW rev.1.

Illustration 9-4: Clearance Dimensions - MR450w Magnet with Roller Dollies Attached

**NOTE:**

* 6 inches (152.4 mm) allowed for roller dollies. Roller dollies taller than 6 inches (152.4 mm) increase overall height.

Minimum height for service clearance is 105 inches (2667 mm). This clearance is needed for Main Lead Extensions, Shim Lead and fill line installation.

Source: 5162096IDW rev.1.

3. Compare the dimensions of the magnet on the moving fixtures being used with the clearances measured along the magnet delivery route. The height can be reduced by putting the moving fixtures directly under the lifting rails. (See [Illustration 9-2.](#))



NOTICE

If the height of the magnet prohibits it from entering a low ceiling area:

- For non-DV systems (HDx, etc.), the 1 in. aluminum spacer plate can be removed from each magnet foot. See [List item To remove the aluminum spacers \(shown below\) from each magnet foot, proceed as follows: Locate a 3/4 in. \(M19\) wrench. Remove the two 3/8 in. hex-head bolts that secure the 1 in. \(25 mm\) aluminum spacer to each foot. Remove and store the aluminum spacers on site.](#)
 - For DV-based systems (MR750, MR450, MR450w, etc.), the 1 in. aluminum spacer plates must remain on the magnet feet during movement to protect the Bridge Vibration Bracket. Contact the Install Base Leader at GEHC Florence for further instruction. The person can be reached at 1-843-667-9799.
4. To remove the aluminum spacers (shown below) from each magnet foot, proceed as follows:
- a. Locate a 3/4 in. (M19) wrench.
 - b. Remove the two 3/8 in. hex-head bolts that secure the 1 in. (25 mm) aluminum spacer to each foot.
 - c. Remove and store the aluminum spacers on site.

Illustration 9-5: Removing Aluminum Spacers



1.2.2 Moving Magnet into MR Suite



CAUTION

Potential Personal Injury

Uneven jacking of the magnet's corners could result in the magnet shifting on the jacks, which may lead to personal injury or magnet damage.

Keep the magnet level at all times during any jacking operation.



NOTICE

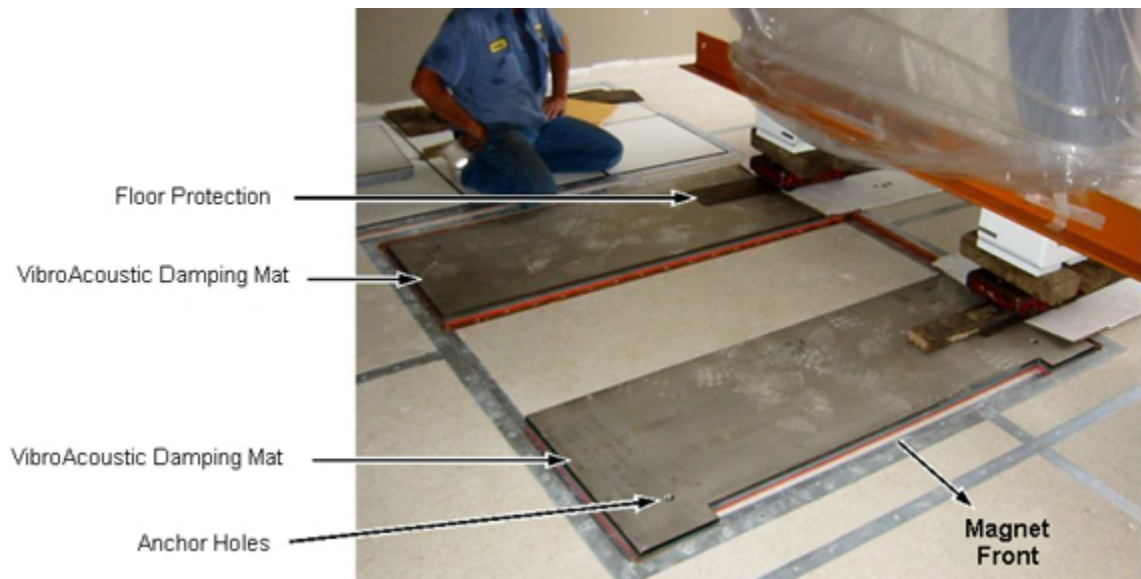
Do not apply any loads to any enclosure cover parts, nor allow straps/cables/chains to scrape enclosure cover parts.

1. If raising the magnet is required, use jacks placed entirely under the lifting rails. (See [Illustration 9-2.](#))
2. Place steel floor plates as needed to protect floors.
3. Move the magnet to the Magnet Room. (If using a motorized tow vehicle, attach cables, chains or straps to the magnet's lifting rails with shackles.)

1.2.3 Moving Magnet Inside MR Suite

1. Cover the Magnet Room floor using steel plates (shown below) from the Magnet Room entrance to the magnet's final position.

Illustration 9-6: Moving Magnet onto Surface Mounted VibroAcoustic Damping Mats



2. Move the magnet to align the holes in the magnet feet to the anchor holes in the VibroAcoustic Damping Mats or the magnet anchors.
3. If VibroAcoustic Damping Mat is used, remove the 1 in. aluminum spacer from the four magnet feet. (See [Illustration 9-5.](#))

4. Verify that the four holes, one per each magnet foot, align (centered ± 0.125 in. or ± 3 mm) over the anchor stud holes in VibroAcoustic Damping Mats.

Illustration 9-7: Magnet Installed on Surface Mounted VibroAcoustic Damping Mats



1.2.4 Lowering Magnet into Position



CAUTION

Potential Personal Injury

Uneven jacking of the magnet's corners could result in the magnet shifting on the jacks, which could lead to personal injury or magnet damage.

Keep the magnet level at all times during any jacking operation.

1. Jack the magnet up sufficiently at the two lifting rails (four corners), and remove the moving fixtures.
2. Slowly lower the magnet onto the Magnet Room floor/spacers/Surface Mounted VibroAcoustic Damping Mats.
3. Release pressure simultaneously in both jacks on one end of the magnet until that end is 1 to 2 in. (25 to 50 mm) lower than the opposite end.
4. Simultaneously lower both jacks on the other end 1 to 2 in. (25 to 50 mm). Repeat lowering the magnet end to end until all feet are on the floor, correctly located on the anchor holes.

Chapter 10 Magnet Leveling, Foot Shimming and Bolt Down

1 Magnet Leveling, Foot Shimming and Bolt Down

1.1 Equipment and Tools

NOTE: This process is for sites without Vibromat. The Vibromat is self-leveling.

- A 3 ft. (1,000 mm) carpenter's level in good condition; no sharp edges/corners, dents, bumps, cracks or loose bubble vial
- Tape measure
- Hammer and wood driving block
- Torque wrench
- Magnet Leveling Kit (46-260888G4) containing the parts listed in [Table 10-1](#).
- Mounting hardware, included in VibroAcoustic Damping Option A (M1060MA):
 - 3 in. x 3 in. (76.8 mm x 76.8 mm) Aluminum Washers
 - 0.75-10 Stainless Steel Studs
 - 0.75-10 Stainless Steel Nuts
 - Stainless Steel Flat Washers

Table 10-1: Magnet Leveling Kit 46-260888G4

Quantity	Part Number	Description
12	2213945	Leveling Shim, 0.062 in. (1.57 mm) thick
8	2213945-2	Leveling Shim, 0.020 in. (0.51 mm) thick
24	2180016	Contact Shim, 6.00 x 6.00 x .020 in. thick (152 x 152 x 0.51 mm)
16	2180016-2	Contact Shim, 6.00 x 6.00 x .032 in. thick (152 x 152 x 0.81 mm)
8	2180016-3	Contact Shim, 6.00 x 6.00 x .040 in. thick (152 x 152 x 1.0 mm)
4	2180016-4	Contact Shim, 6.00 x 6.00 x .063 in. thick (152 x 152 x 1.6 mm)

1.2 Height and Level Adjustment



NOTICE

Magnet and dock installation and placement are critical to image quality and hardware reliability. Magnet must be installed level with iso-center to specifications listed in [Section 1.2.2](#).



NOTICE

Make sure the magnet height stays within specification while leveling the magnet.

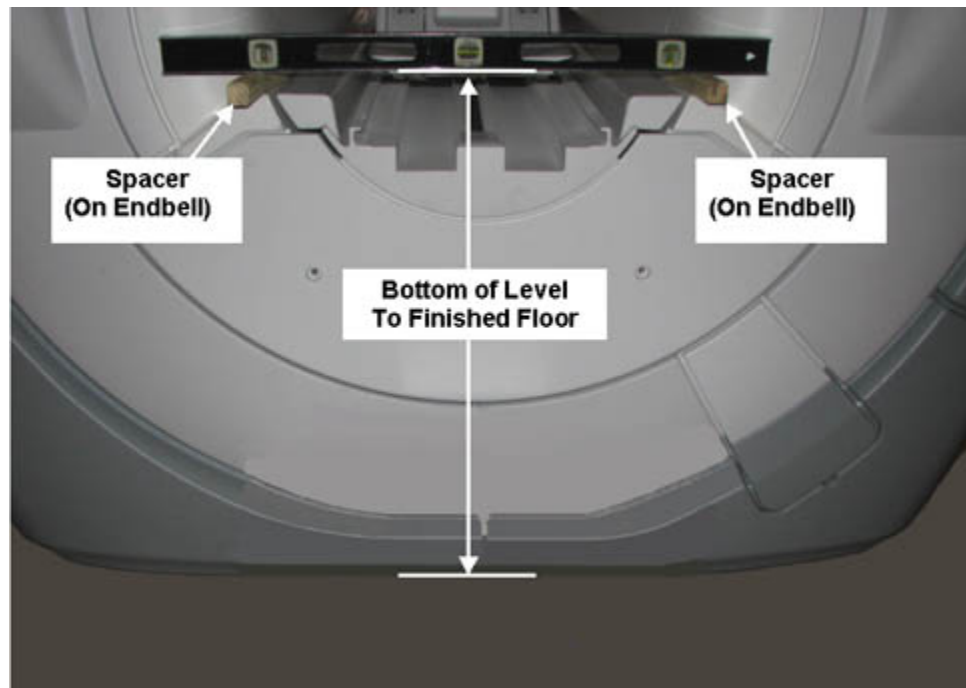
1.2.1 Magnet Leveling - Left to Right



NOTICE

The magnet level measurement must be taken from the magnet endbell, not the bridge. The bridge is higher than the endbell. Place a spacer of the same size between the level and magnet endbell from left to right. [Illustration 10-1](#) shows two wood blocks as spacers that are both 1.25 inch (33 mm) high.

Illustration 10-1: Magnet Leveling - Left to Right



1. Hold a 3 ft. (1,000 mm) carpenter's level across the flats of the magnet front endbell.
2. Level magnet left to right.

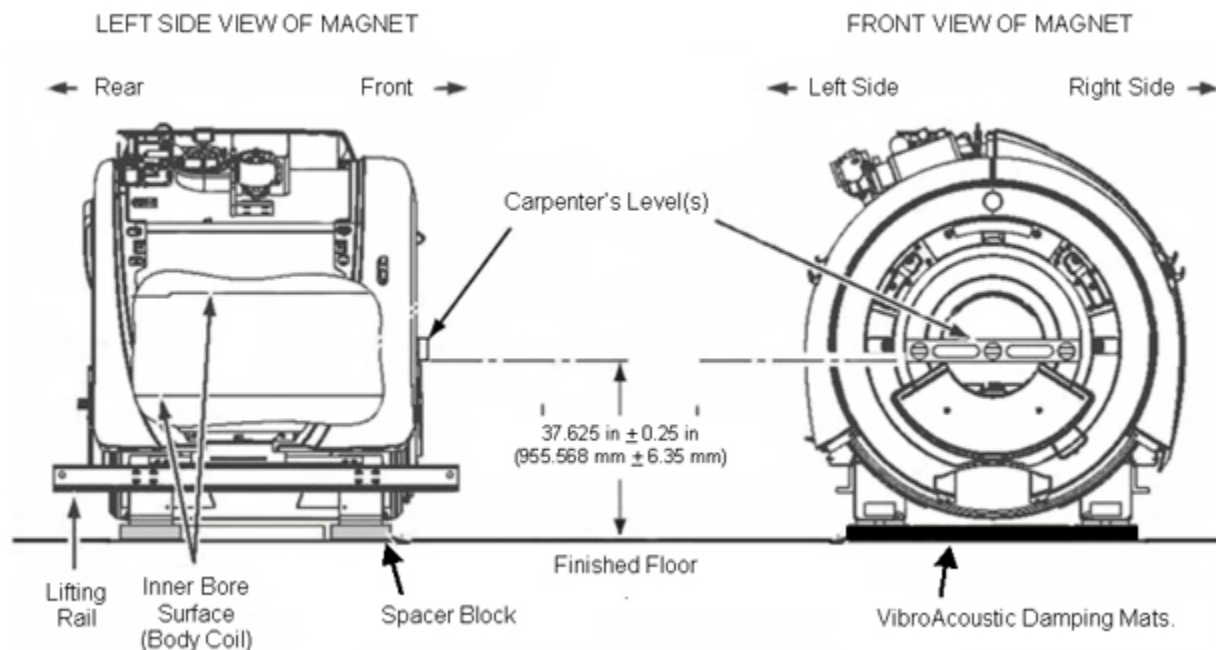
1.2.2 Measuring Magnet Height

1. The magnet level measurement must be taken from the magnet endbell (not the bridge) to the finished floor. The bridge is above the magnet endbell. Place a spacer of the same size between the level and magnet endbell from left to right. [Illustration 10-1](#) shows two wood blocks as spacers that are both 1.25 inch (33 mm) high.

NOTE: The magnet level measurement must be taken from the magnet endbell, not the bridge.

2. Measure the vertical distance from the **bottom** of the carpenter's level to the **finished** floor. (See [Illustration 10-2](#).) The final installation specification measurement for LCC magnets (HDx, HDxt, and MR750) should be 37.625 in. $\pm$ 0.25 in. (955.675 mm $\pm$ 6.35 mm). In the example in [Illustration 10-1](#), spacer height + magnet specification = measured height (bottom of level to finished floor) or 1.25 inch + 3.675 inch = 37.625 inches.

Illustration 10-2: Measuring Magnet Height and Checking Magnet Levelness



3. Add leveling shims under each magnet foot according to the next section, Checking Magnet Levelness, to achieve the correct height.

1.2.3 Magnet Leveling - Front to Back

1. Position a 3 ft. (1,000 mm) carpenter's level at the top of the RF coil. (See [Illustration 10-3.](#)) Make sure the entire length of the level is on the RF coil, it should not be on the endbells.

NOTE: The use of other locations to measure magnet height is not recommended due to manufacturing tolerance issues.

Illustration 10-3: Magnet Leveling - Front to Back



2. If the level's bubble is not between the scribe lines:
 - a. Use a jack to raise the low magnet foot.
NOTE: Jacking the lift rail is covered in [Chapter 9, Moving Magnet to MR Suite](#).
 - b. Insert the appropriate thickness of aluminum shim plates under the low magnet foot.
 - c. Recheck side-to-side levelness.
 - d. Recheck magnet height to conform with [Section 1.2.2](#).
 - e. Repeat Steps b through d until the front of the magnet is level side-to-side and within the correct height specification.

1.3 Magnet Anchoring



NOTICE

Complete contact between the bottom of the magnet feet and the floor or Vibroacoustic mat is important to minimize magnet motion/vibration, which cause image problems.



CAUTION

Potential Personal Injury

Contact shims have sharp edges that can cause personal injury.

Always tape shim edges to the floor surface.

1. After leveling is completed, anchor the magnet to the floor or Vibroacoustic mat . (See [Illustration 10-4.](#))
2. Make sure all nuts are properly torqued.

Immediately continue with the *Conversion to Operating Configuration* and *Exhaust Gas Vent Connection* subsections of the Set-Up and Calibration chapter in the appropriate manual:

- For HDi, HDx and HDxt: Direction 2325852, *LCC300 Active Shield Magnet and Cryogenics Subsystem*.
- For MR750: Direction 5500099, *MR750 3.0T LCC300 Active Shield Magnet and Cryogenics Subsystem*.



NOTICE

The magnet **MUST** be connected to the helium exhaust gas vent within 24 hours.

Illustration 10-4: Magnet Anchoring

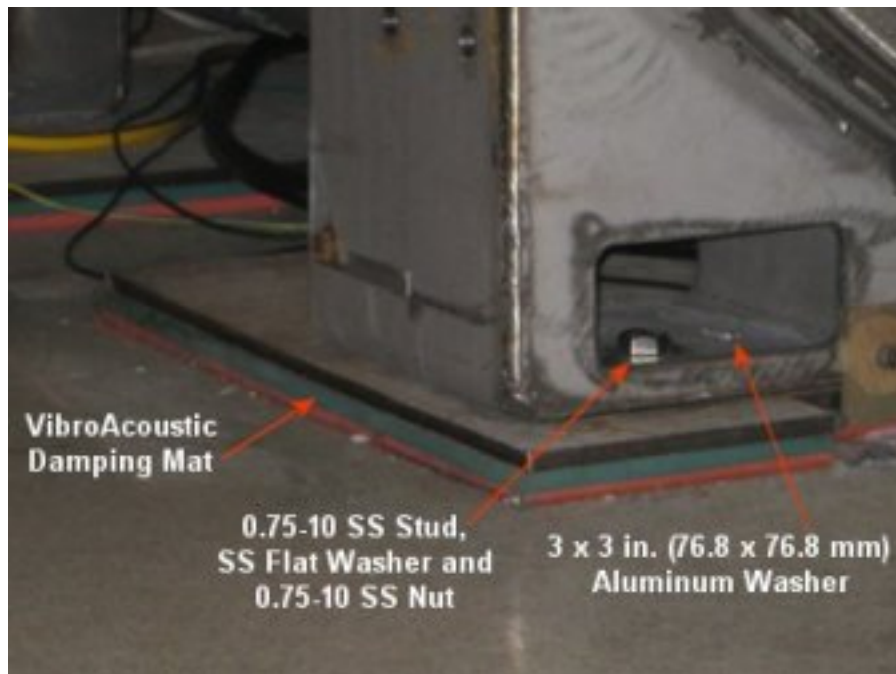
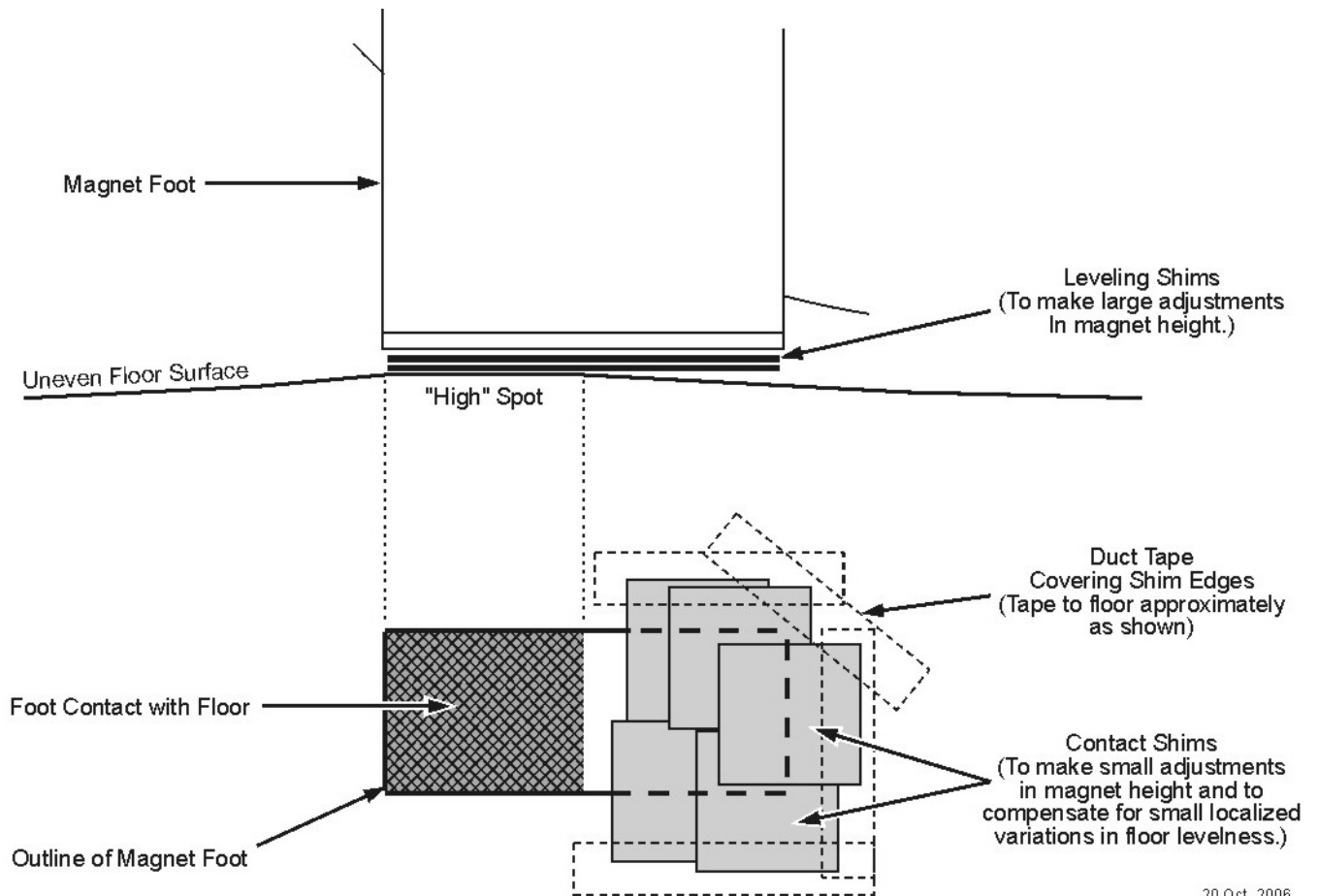


Illustration 10-5: Typical Shim Arrangement for Gap Fill



20 Oct. 2006

**! WARNING****ASPHYXIATION HAZARD!**

THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES IF EITHER THE LIQUID HELIUM LEVEL IS TOO LOW OR THE HELIUM VESSEL TEMPERATURE LEVEL IS DISRUPTED.

MAGNET ROOM EXHAUST GAS VENT MUST BE CONNECTED WITHIN 24 HOURS OF MOVING THE MAGNET INTO THE MAGNET ROOM.

Chapter 11 Dock Assembly Floor Anchor Installation

1 Dock Assembly Floor Anchor Installation

1.1 Attachment to Magnet



WARNING

POTENTIAL PROJECTILE HAZARD!
DOCK CONTAINS FERROMAGNETIC MATERIAL.

DO NOT INSTALL OR ADJUST DOCK INSTALLATION WHILE MAGNET IS RAMPED.

1. For HDx-based systems, fasten the Dock Mounting Support (P/N 2226264) to the magnet using the 11 mm inside diameter washers and the hex-head screws supplied with the support; apply Loctite® 242 as required. (See [Illustration 11-1](#).) For Discovery-based systems, the Bridge Mounting Support is shipped attached to the magnet by phenolic washers and hex-head screws supplied with the support. (See [Illustration 11-2](#).)
2. For HDx-based systems, slip two washers onto each of the two middle studs on the Mounting Support as shown in [Illustration 11-1](#). For Discovery-based systems, slip two washers onto each of the four middle stud on the Mounting Support as shown in [Illustration 11-2](#) (only two washers shown).

NOTE: Add or remove washers between the Mounting Support and Dock Assembly if alignment adjustments are required when installing the table.

Illustration 11-1: Attaching Mounting Support & Dock Assembly to Magnet (HDx-based Systems)

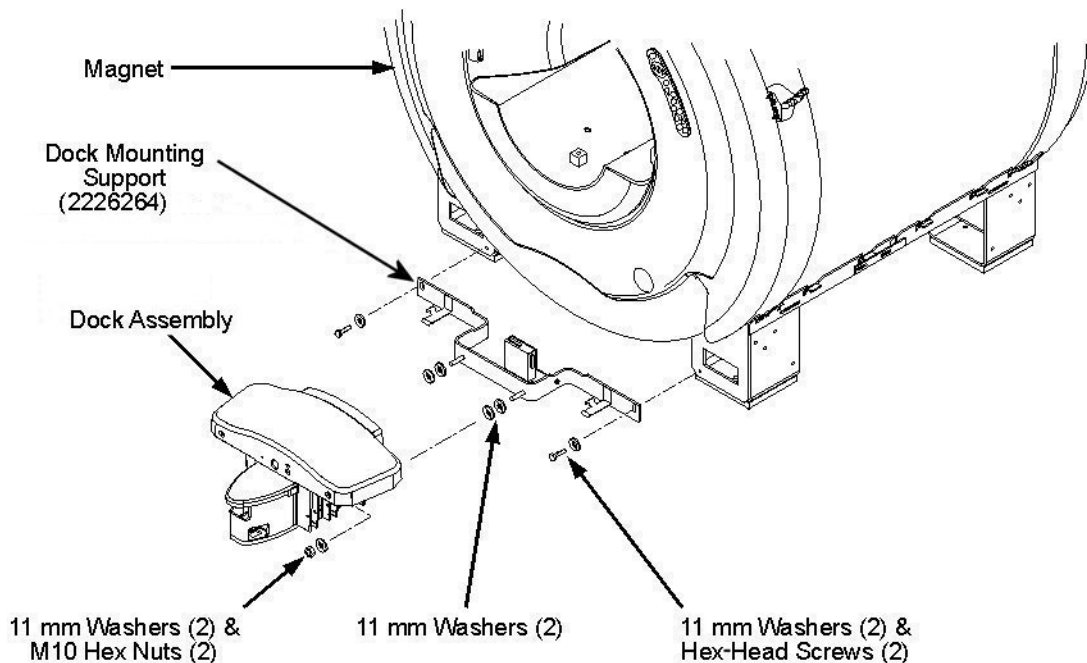
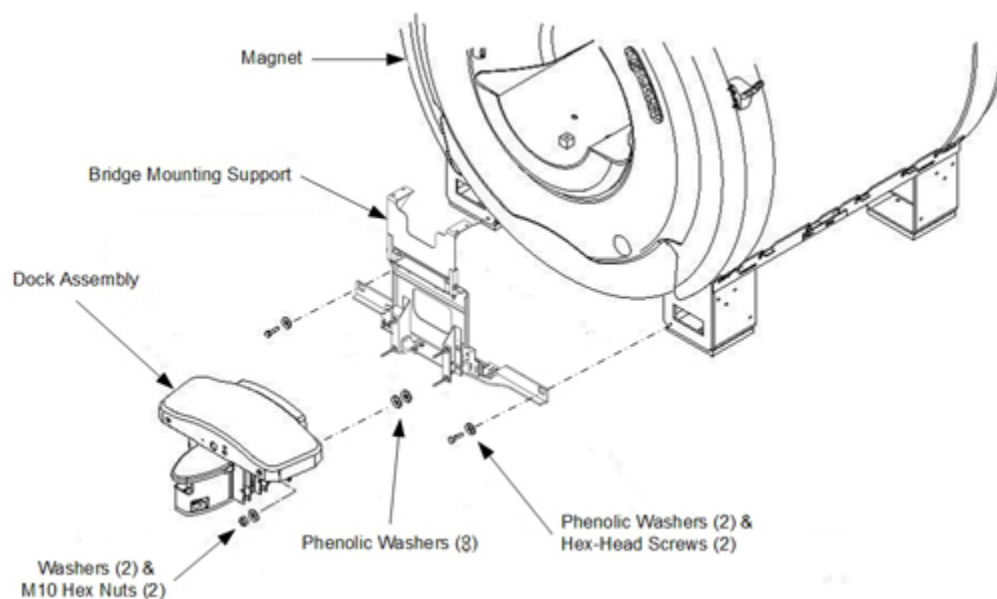


Illustration 11-2: Attaching Mounting Support & Dock Assembly to Magnet (Discovery-based Systems)

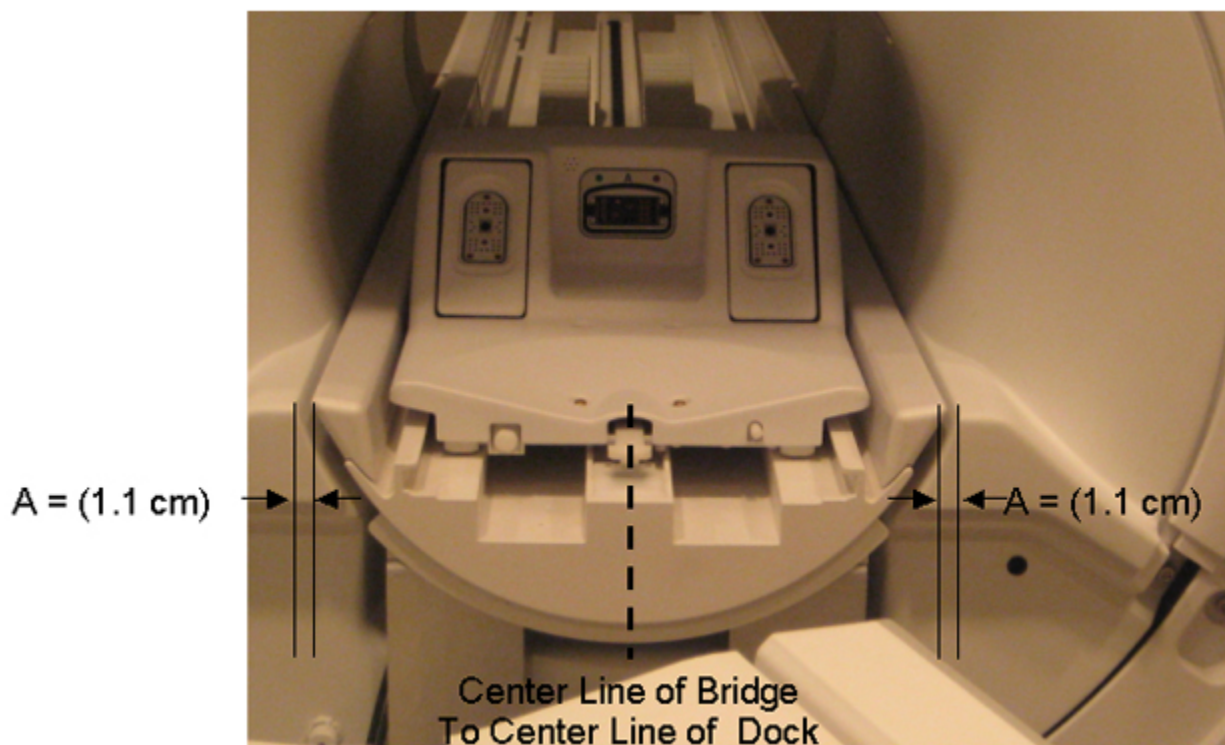


3. Verify gap between bridge and endbell is the same on both sides (approximately 1.1 cm for Discovery/Optima-based systems). Adjust as needed by loosening phenolic washers located on magnet feet.

NOTE: If adjusting left to right alignment, be sure to maintain a gap between the bottom surface of the dock mounting support and magnet room floor.

4. Locate center line of bridge for proper alignment of dock anchor. (See [Illustration 11-3.](#))

Illustration 11-3: Locating Bridge Left to Right Center Line



5. Slide the Dock Assembly onto the Mounting Support studs. Be sure to route cable down the center of the dock support bracket as shown in [Illustration 11-4](#)

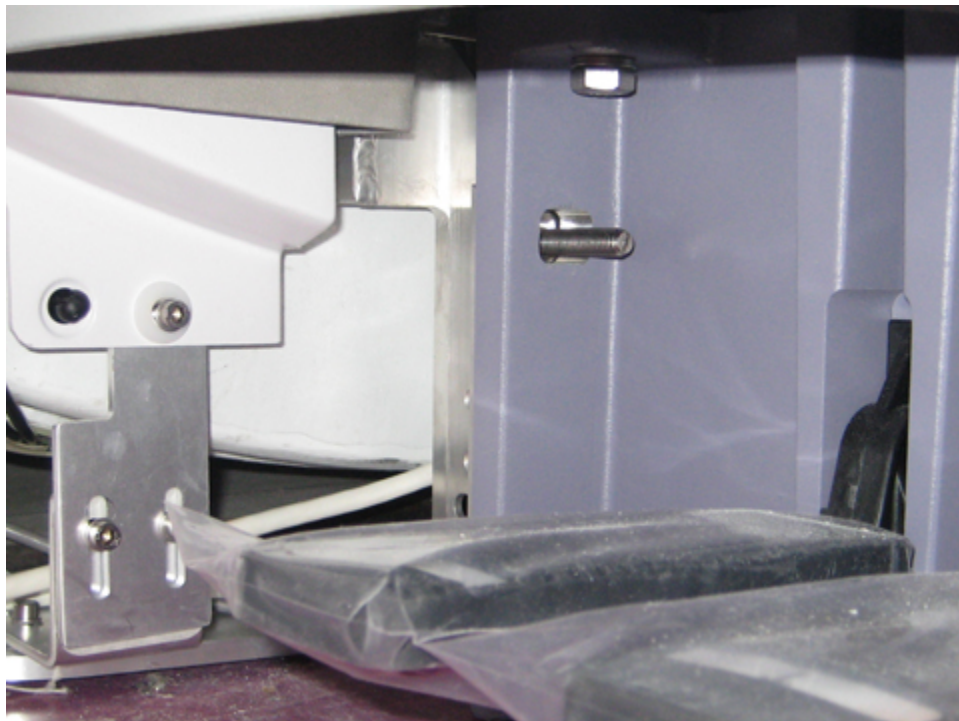
NOTE: For Discovery-based systems, if the Dock Mounting Support studs are not installed, install them at this time by applying a small amount of Loctite to the threaded rod and secure with a lock washer on the back side of the lower support bracket.

Illustration 11-4: Dock Cable Routing



6. Adjust the dock left to right in order to center the threaded rods to the dock mounting holes. (See [Illustration 11-5.](#))

Illustration 11-5: Dock Alignment - Left to Right



7. Secure the Dock Assembly to each stud on the Mounting Support using the two washers and hex nuts. (See [Illustration 11-1](#).)
8. For Discovery/Optima-based systems, verify that there is no contact between the bridge support bracket and the front of the magnet when the front endbell and bridge surfaces are flush (in/out). (See [Illustration 11-6](#).)

Illustration 11-6: Location of Bridge Support Bracket to Magnet



1.2 Attachment to Floor

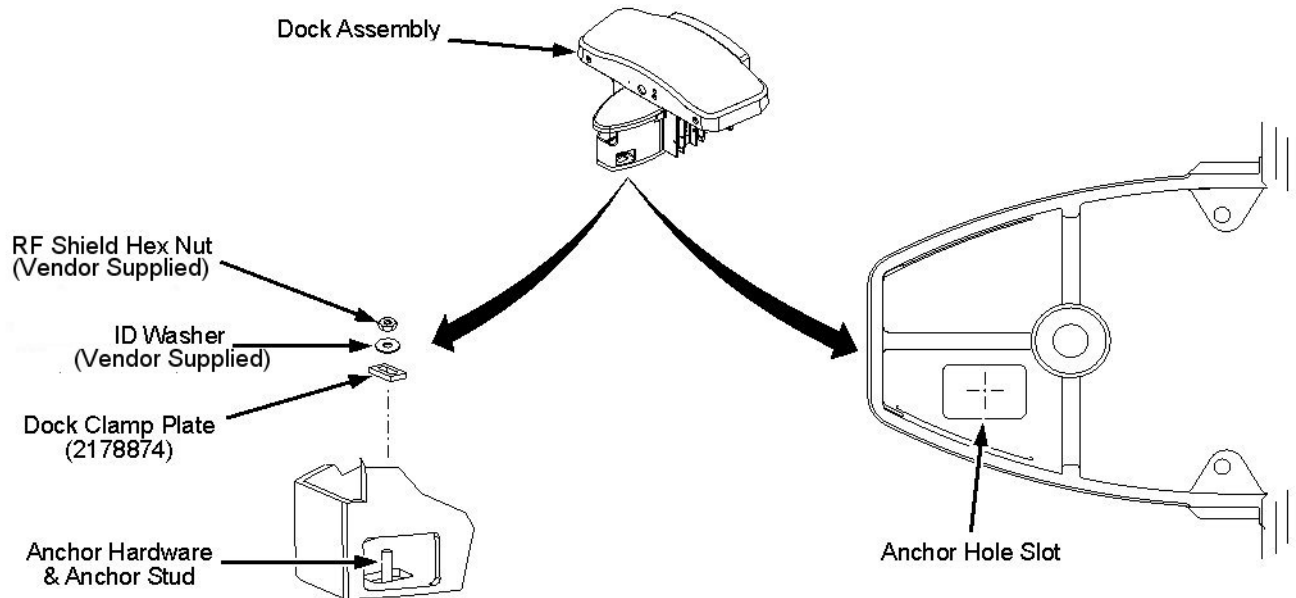


NOTICE

Make sure the floor anchor does NOT short to ground.

1. Using the Dock Assembly as a template, mark the anchor location in the center of the anchor hole slot. (See [Illustration 11-7](#))
2. Unfasten the hex nuts and washers installed in [Section 1.1](#), Step 5, and remove the Dock Assembly.

Illustration 11-7: Attaching Mounting Support & Dock Assembly to Floor



NOTE: The floor anchor stud installation in the following step is to be done by the **RF Room Vendor/Mechanical Contractor**.

3. **RF Room Vendor/Mechanical Contractor** to drill a hole in the floor and install anchor hardware and stud, and tighten the stud to torque specification per the pre-installation manual.
4. **RF Room Vendor/Mechanical Contractor** to verify that the anchor is not shorted to ground; make corrections as necessary.
5. Reinstall the Dock Assembly using the vendor-supplied hex nuts and washers; apply Loctite® 242 as necessary.
6. Fasten the Dock Assembly to the floor using a Dock Clamp Plate (2178874) and the vendor-supplied hex nut and ID washer; apply Loctite® 242 as required.
7. Make any final adjustments needed when the table is installed.

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